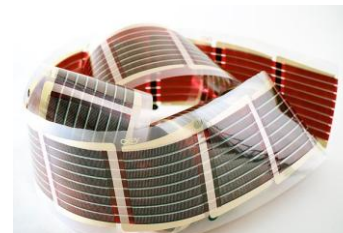
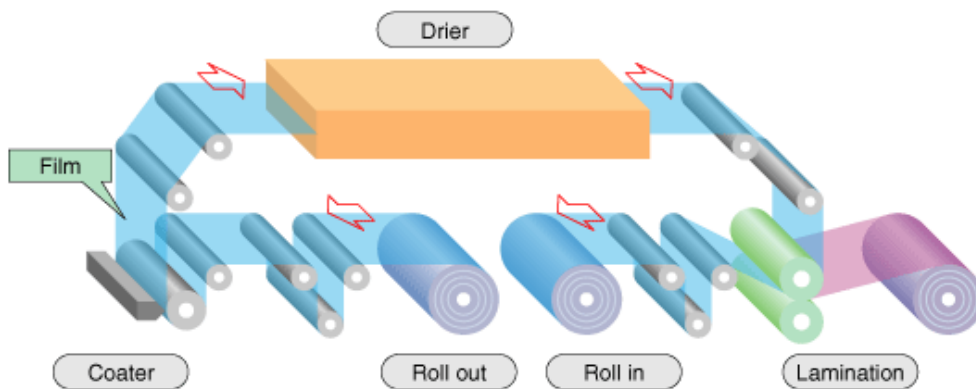


Fundamentals of Photovoltaics (EE336)

Lecture 7 – Organic solar cell

Organic Solar Cell

- Organic Solar Cell (OSC) or Organic Photovoltaics (OPV)
- Photovoltaic devices employing molecular materials (organic semiconductors), including **conjugated polymers** and **small molecules**.
- Advantages:
 - solution-processing at room temperature
 - allow low-cost and large-area production (e.g., roll-to-roll manufacturing process)
 - Light-weight and flexible

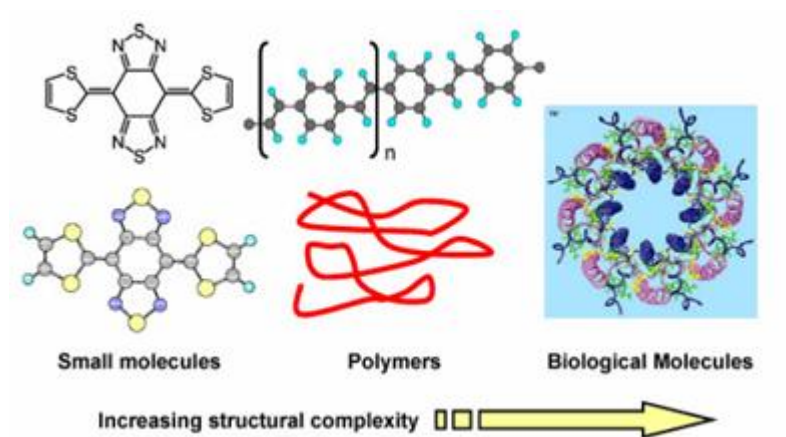


Organic Semiconductors

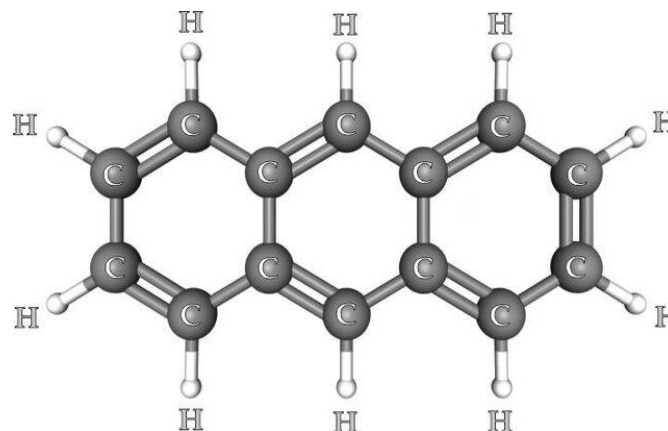
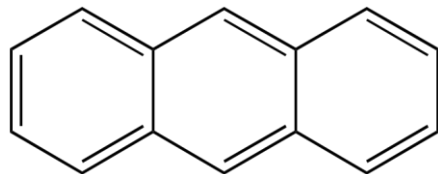
- Based on the type of bonding interactions between the constituent atoms and molecules, solids are traditionally classified as covalent, ionic, metallic, or molecular.
- Although some organic solids exhibit metallic or ionic characteristics, the majority of organic solids belong to the category of **molecular solids**.
- Each molecule has a stable structure due to the strong covalent bonding between atoms.
- But the intermolecular interaction forces that hold the molecules together is primarily of the **van der Waals** type, which is considerably weaker than that of covalent or ionic bonds.
- Hence, organic materials differ significantly in their optical, mechanical, and electronic properties from the conventional semiconductors.

Organic Semiconductors

- Organic materials can be further divided into three types.
- With increasing structural complexity and molecular weight, there are small molecular weight compounds, polymers, and biological molecules.
- Small molecular weight compounds are often referred as “small molecules”, consisting of up to a few hundred atoms, and having molecular weights $MW < 1000$.
- Polymers are long-chain macro-molecules consisting of an undefined number of repeating units; therefore, their MW can be anywhere from a few thousand to more than a million.
- The molecules having the highest structural complexity are of biological origin, such as DNA.
- The term organic electronic materials or organic semiconductors usually refer to small molecules and polymers.

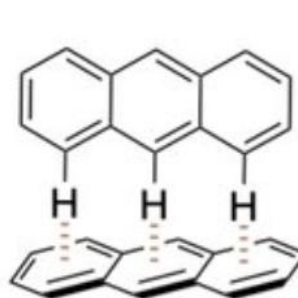
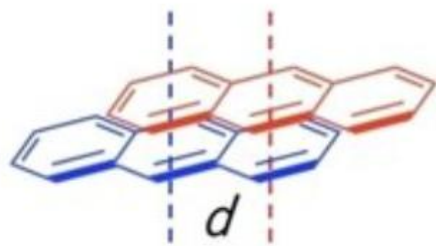


Anthracene (Small Molecule)

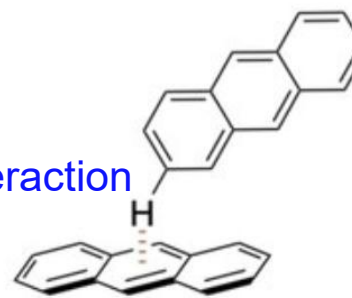


Intramolecular interaction (within a single molecule)
e.g, C-H bond, covalent bond (strong)

π - π interaction



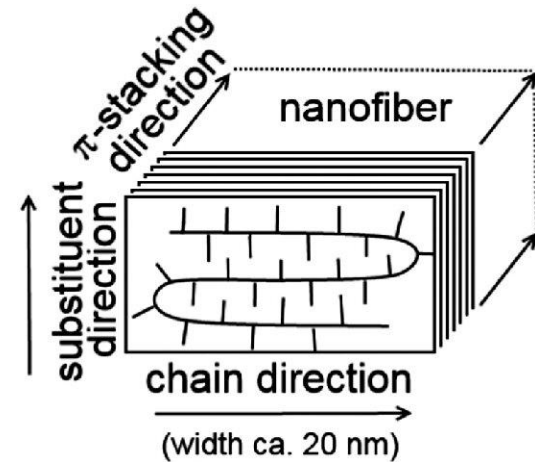
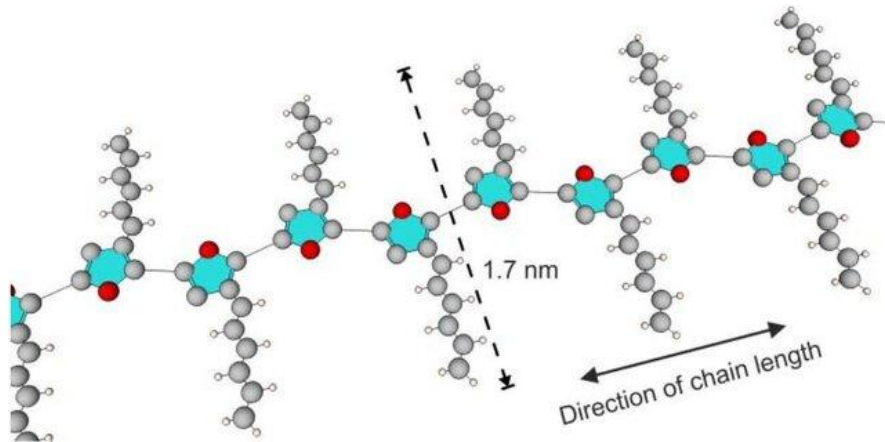
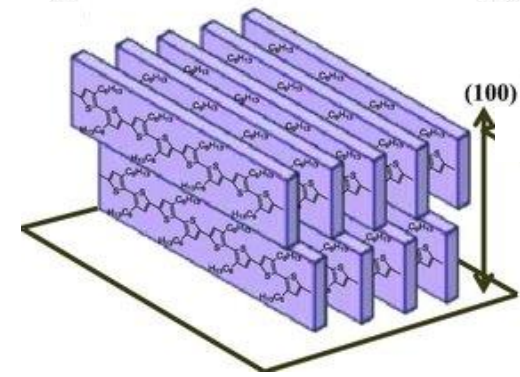
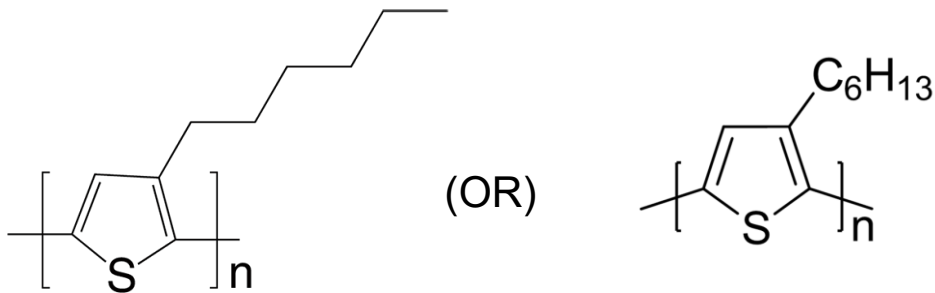
CH - π interaction



Intermolecular interaction (between molecules) e.g, CH - π interaction, π - π interaction, Van der Waals interaction (weak)

<https://stock.adobe.com/video/3d-structure-of-anthracene-3d-view-of-molecular-structure/1011634133>

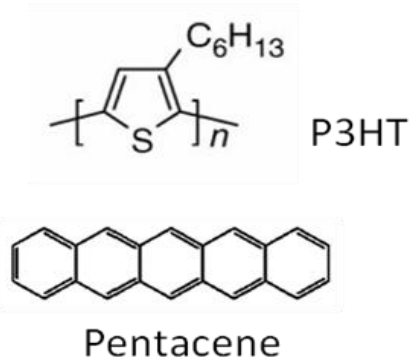
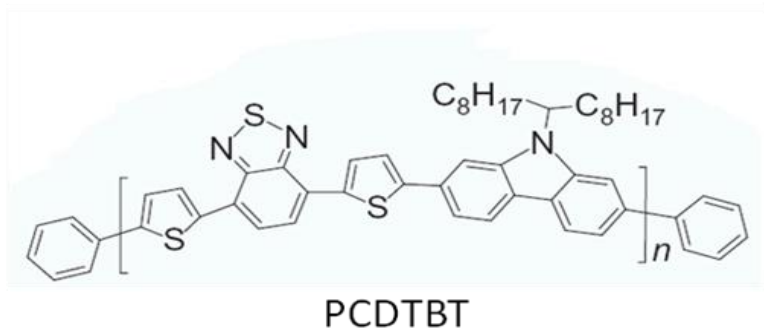
Poly(3-hexylthiophene-2,5-diyl), P3HT (Polymer)



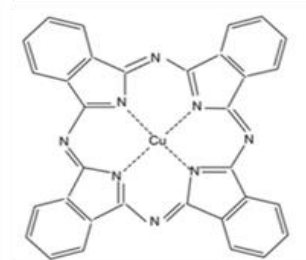
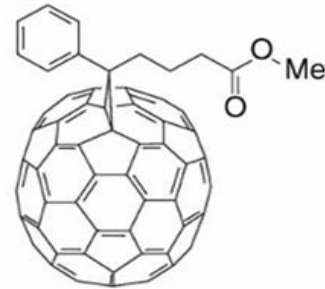
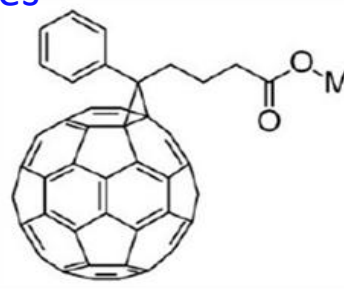
Intramolecular interaction (within the chain segment)
e.g, C-H bond, C-C bond, covalent bond (strong)

Intermolecular interaction (between the chain segments)
e.g, $\pi - \pi$ interaction, Van der Waals interaction (weak)

Donors



Fullerene derivatives

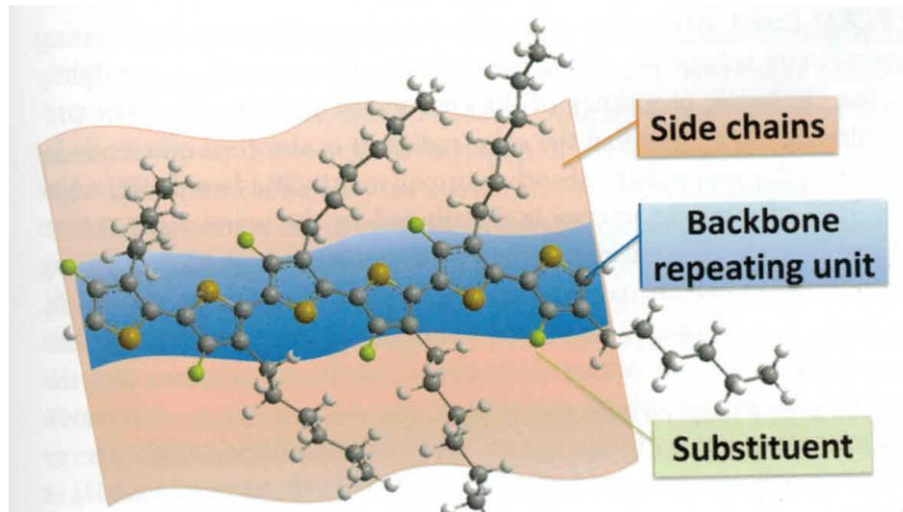


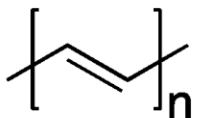
Acceptors

Chemical structures of some typical molecular semiconductors used in OPVs

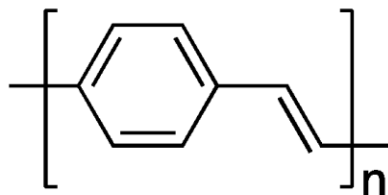
Structural features of conjugated polymers

- A conjugated polymer (CP) can be arbitrarily divided into three constituting components: **conjugated backbone**, the **side chains**, and the **substituents**.
- The conjugated backbone rules most of the solar cell-related physical properties of the CP, such as energy level, bandgap and intermolecular interactions.
- Side chain determines the solubility to the CP, which plays a crucial role in improving the molecular weight and processability. Improper attachment of side chains can introduce twisting in the CP backbone, leading to a larger bandgap, lower carrier mobility and poor PV properties.
- Substituents is used as fine-tuning method to tweak the electronic properties of CP (e.g., energy levels, bandgap and mobility)

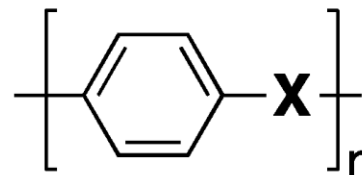




polyacetylene

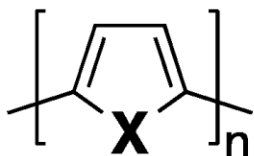


polyphenylene vinylene



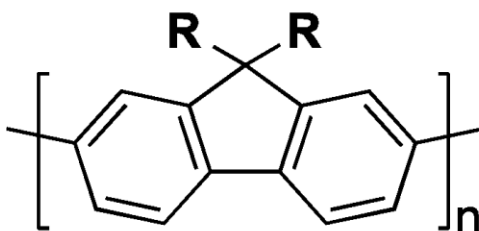
X: NH/H, polyaniline

X: S, polyphenylene sulfide

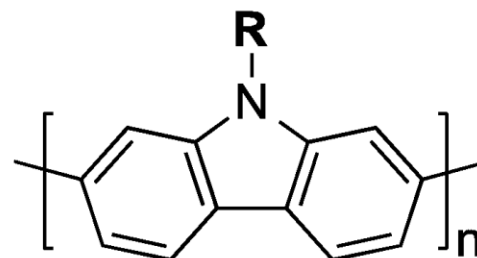


X: NH, polypyrrole

X: S, polythiophene



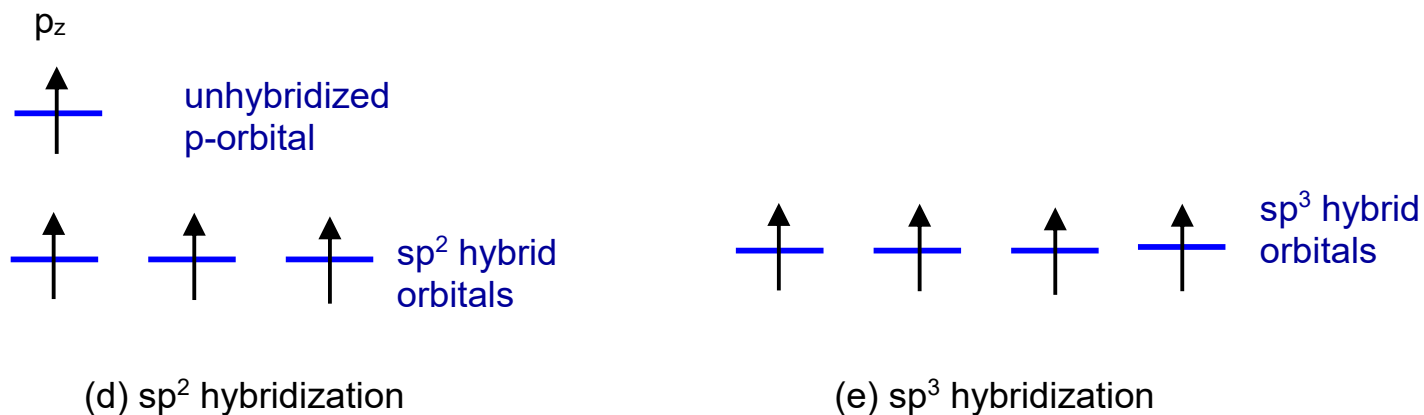
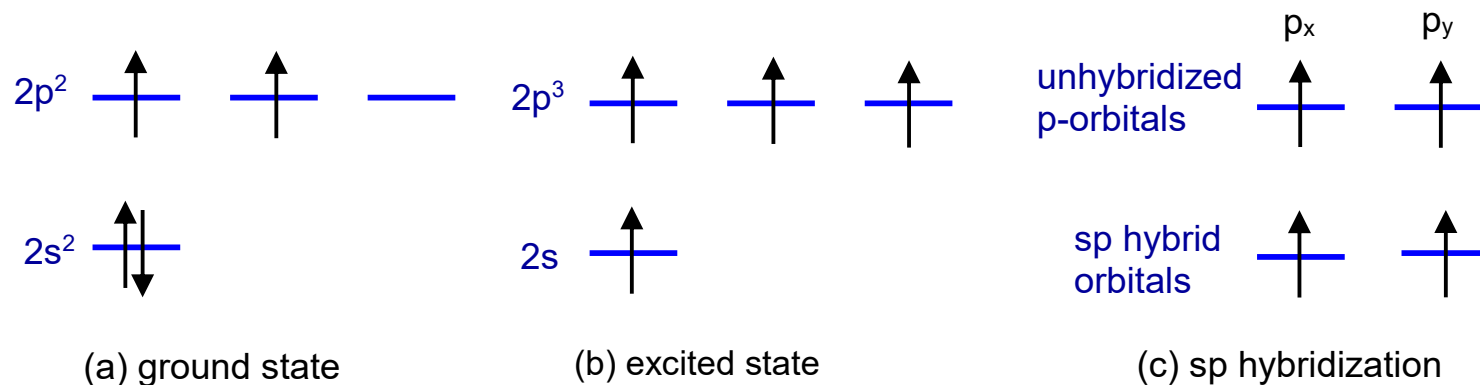
polyfluorene

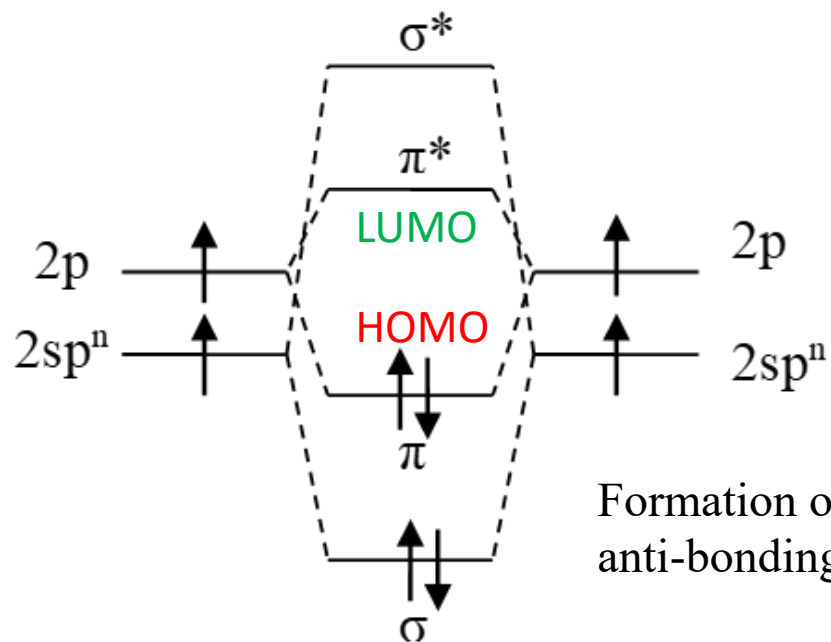


polycarbazole

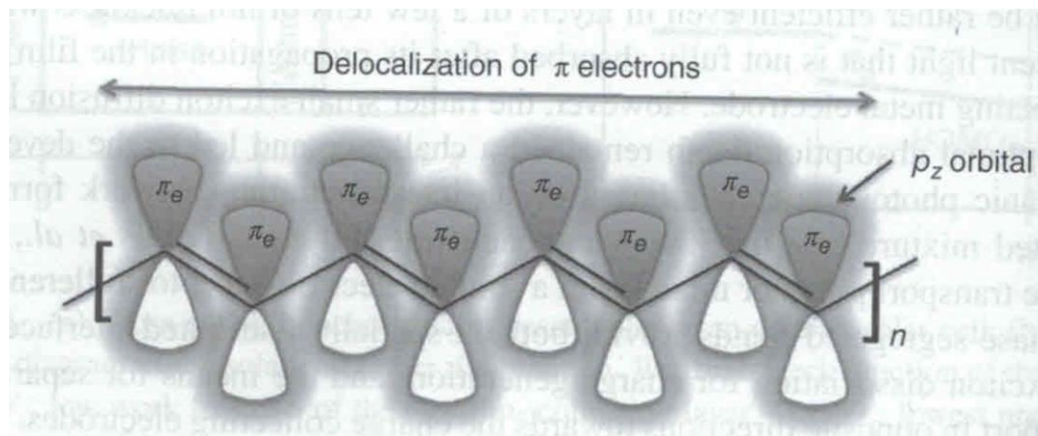
Electronic Structure of organic semiconductor

- The electronic configuration of a carbon atom in the ground state is $1s^2 2s^2 2p^2$
- Two of the four valence electrons are paired and occupy the 2s atomic orbital, whereas the other two are unpaired, and each occupies one of the three degenerate 2p orbitals.
- Due to the Pauli Exclusion Principle, only the unpaired 2p electrons in this configuration may be shared with other atoms to form covalent bonds.
- The fully occupied 2s orbital, however, may excite and mix or hybridize with one or more 2p orbitals, leading to single occupancy on each of the four orbitals.
- The hybridized configurations are more favorable than the ground state for the carbon atoms to form covalent bonds with other atoms, as all four valence electrons are now available for bonding.





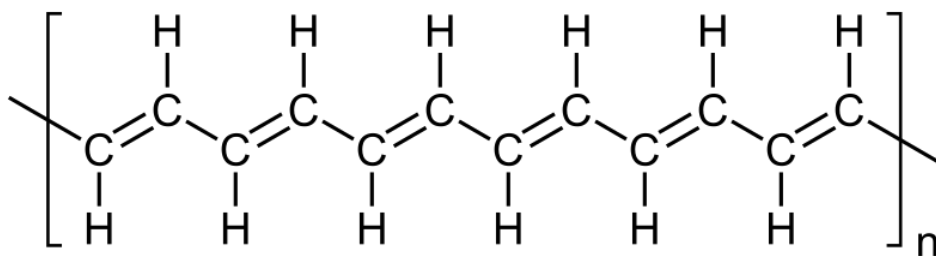
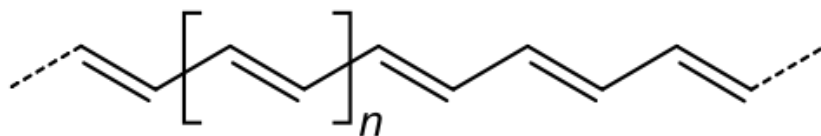
Formation of molecular orbitals (MO) and their anti-bonding MOs from two adjacent atoms



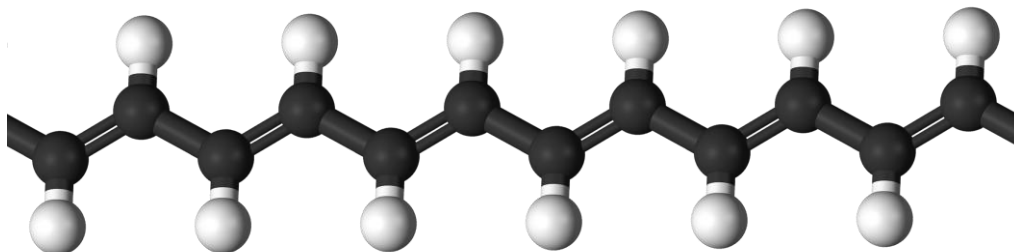
Electronic structure of organic semiconductor

- The molecular structure of organic semiconductors presents a backbone along which the carbon (or nitrogen, oxygen or sulfur) atoms are sp^2 -hybridized and thus possesses a π -atomic orbital.
- The conjugation (overlap) of these π orbitals along the backbone results in the formation of delocalized π molecular orbitals.
- The overlap of different electron wave functions of neighbouring atoms defines the frontier electronic levels: the Highest Occupied Molecular Orbital (HOMO) and the Lowest Unoccupied Molecular Orbital (LUMO).
- HOMO and LUMO determine the optical and electrical properties of the organic semiconductors.
- The **overlap of the π molecular orbitals** between adjacent molecules or polymer chains characterizes the strength of **the intermolecular electronic couplings**.

Polyacetylene



Structural diagram



Ball-and-stick model

The Nobel Prize in Chemistry 2000



Alan J. Heeger
Prize share: 1/3



Alan G. MacDiarmid
Prize share: 1/3



Hideki Shirakawa
Prize share: 1/3

The Nobel Prize in Chemistry 2000 was awarded jointly to Alan J. Heeger, Alan G. MacDiarmid and Hideki Shirakawa *"for the discovery and development of conductive polymers"*.

Photos: Copyright © The Nobel Foundation

Organic semiconductor

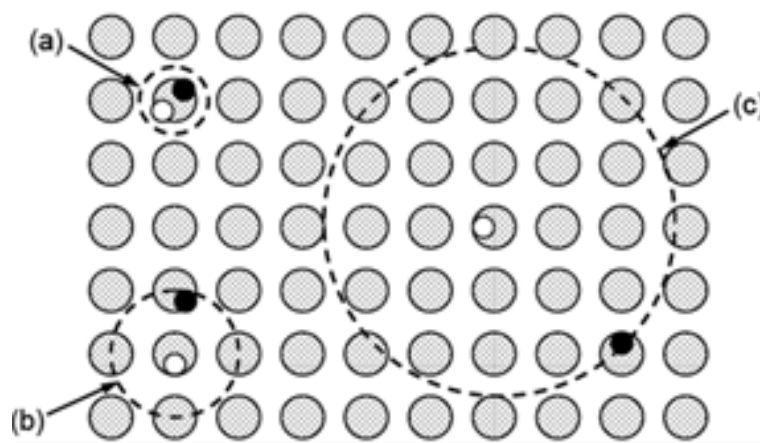
- In crystalline inorganic semiconductor, the three-dimensional character and rigidity of the lattice ensure wide valence and conduction bands and large charge carrier mobility values (10^2 to 10^3 cm²/Vs)
- In organic semiconductors, the **weakness of the electronic couplings** (due to their intermolecular character) and **disorder effects** produce low carrier mobility values due to **charge-carrier localization** and **formation of polarons**.
- Thus, transport in organic semiconductor relies on **polaron-hopping** from molecule to molecule or between polymer chain segments.
- As a result, charge carrier mobility values strongly depend on the morphology and can vary over several orders of magnitude.
- Highly disordered amorphous film: 10^{-6} to 10^{-3} cm²/Vs
- Highly ordered crystalline film: > 1 cm²/Vs
- One challenge in OPV is to design high-mobility polymers and molecules ($> 10^{-2}$ cm²/Vs) so that carrier diffusion lengths are of the order of the thickness of the light absorber (typically 100-200 nm)

Organic semiconductor

- Absorption coefficients of organic semiconductors are usually one order of magnitude higher than inorganic semiconductors.
- The **high extinction coefficients** are a result of the large wave function overlap between the ground state and the lowest excited states.
- After excitation, the π -conjugated system relaxes down to the bottom of the potential energy of the lowest excited state (excited state reaches its equilibrium geometry), and an exciton forms.
- The **binding energy of an exciton** in the organic semiconductor is on the order of **a few tenths of an eV**, much higher than thermal energy.
- This high value is due to a **low dielectric constant** of π -conjugated organic materials ($\epsilon_r = 3 - 4$) and the significant **electron correlation and geometry relaxation effect** in these materials.
- Thus, in contrast to inorganic semiconductors, the absorption of a photon at room temperature in conjugated materials does not lead to free charge carriers but to neutral, bound electron-hole pairs (excitons)
- Therefore, the dissociation of these excitons into mobile free carriers plays a central role in the design and architecture of OPV.

Exciton in Organic Semiconductor

- An exciton is the photoexcited state of an electron-hole pair bound together by the Coulomb attraction.
- Excitons are broadly classified in three classes depending on the strength of the **Coulomb attraction**. They are: (a) the Frenkel exciton, (b) the charge-transfer exciton (CT) and (c) the Mott-Wannier exciton



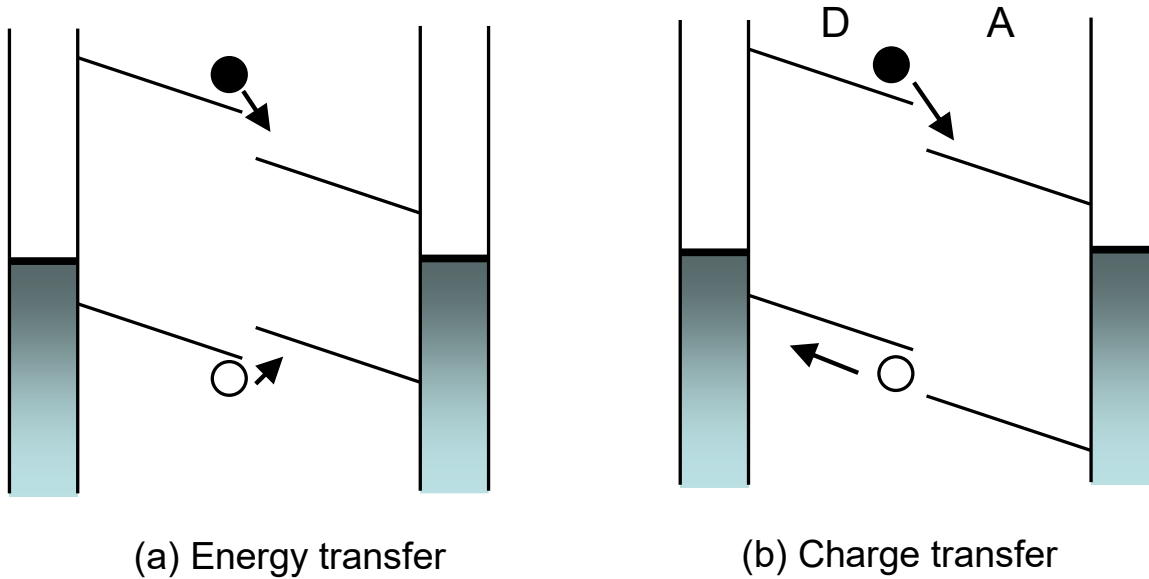
Exciton in Organic Semiconductor

- A Frenkel exciton corresponds to a molecular excited state where electron and hole reside on the same molecule. The radius of the exciton, R , is less than or close to the lattice constant a .
- With increasing intermolecular interactions, the excitonic state may delocalize over two or more adjacent molecules, forming a charge-transfer (CT) exciton.
- While Frenkel and CT excitons are present in organic semiconductors, highly delocalized, hydrogen-like Mott-Wannier excitons with $R \gg a$ are commonly found in inorganic semiconductors.
- Due to the Coulombic interaction between the constituent electron and hole, the Frenkel or CT excitons are tightly bound, with a binding energy ranging from 0.1 eV to 2 eV, while that for Mott-Wannier excitons in inorganic semiconductors is only a few *meV*.
- The Frenkel exciton plays a dominant role in organic π -conjugated molecules, whereas the CT exciton is important in the case of molecular solids such as anthracene.

Exciton in Organic Semiconductor

- The motion of excitons in organic semiconductors can be diffusion among molecules and later charge transfer or energy transfer between different species of molecules. Similar to the transport of single carriers, exciton diffusion in organic semiconductors can occur through **band transport** or via the **hopping process**.
- Energy transfer between a molecule (initial exciton site) and another molecule (final exciton site) may occur through **Forster energy transfer**.
- An exciton that carries its excitation energy can be transferred from the material with the higher bandgap to the other if both the hole and electron can decrease their energy thereby. This process of excitation energy transfer which is accompanied with some energy loss is also known as *Foster energy transfer* and can be used to shift the emission band of the exciton towards lower energies in LEDs. Foster energy transfer also allows the design of a possibility helpful directionality in an OPV since it is a one way mechanism, carrying excitons from the shorter wavelength absorber to the longer wavelength absorber.

Exciton in Organic Semiconductor



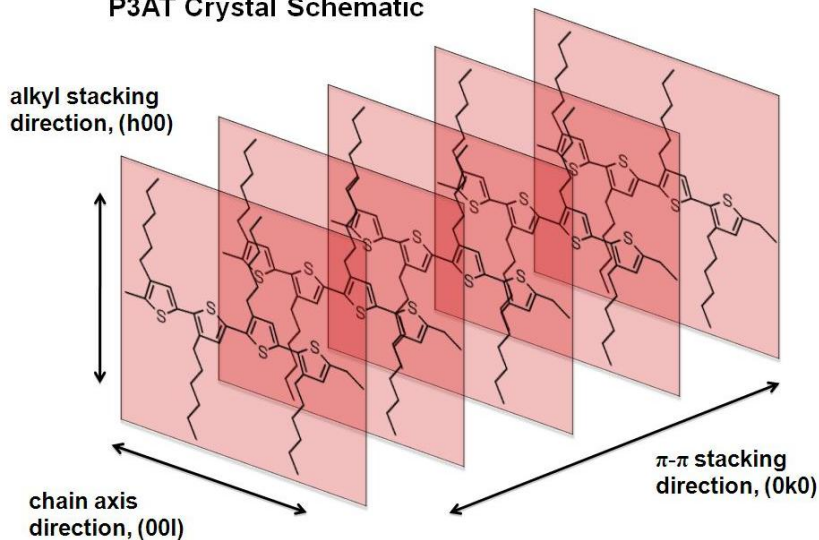
Exciton in Organic Semiconductor

- Beside energy transfer to another molecule, charge transfer may occur, and excitons may also dissociate into a free electron and hole when it is subject to an electric field or at hetero-interfaces.
- Since the binding energy of exciton in inorganic semiconductor is very small, a built-in electric field at p-n junction or even thermal energy at room temperature is able to dissociate the exciton.
- However, binding energy of exciton in organic material is so large that an intense electric field is needed to cause Frenkel excitons to dissociate.
- One of the most effective means to dissociate an exciton is to use a **donor/acceptor (D/A) hetero-interface** with large enough offset between LUMOs of D and A. The band offset between HOMOs of D and A molecules and/or that between their LUMOs provides excess energy for the interfacial dissociation of an exciton. Typically, **about 0.4 eV offset** is required to overcome the exciton binding energy of 0.4 eV. Otherwise charge transfer may occur but the excitons do not split into their constituent charges and recombine eventually at the D/A interface. Hence a D/A hetero-interface must be effectively optimized to improve the efficiency of OPV.

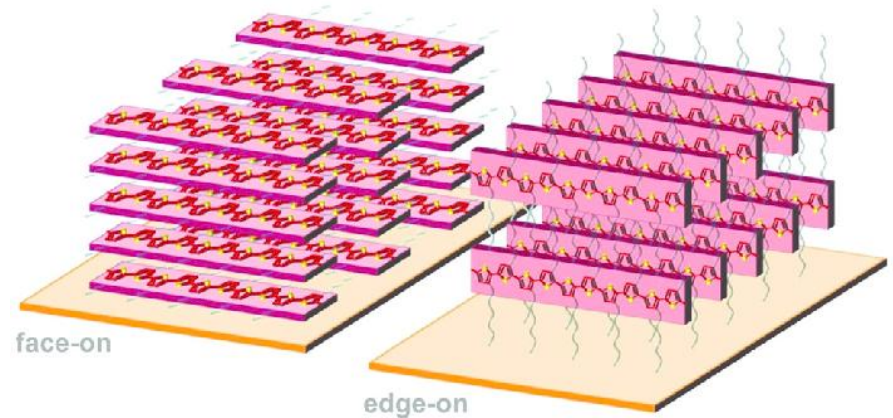
Charge transport and molecular orientation

- Transport of carriers is limited by “hopping” between molecules.
- Since organic molecules (and particularly polymers) typically exhibit low symmetry, transport can be highly **anisotropic**.
- Intra-chain (intramolecular) transport will be faster than transport in the π – π stacking direction, while lateral transport through saturated side-groups can be very poor.

P3AT Crystal Schematic



Further Reading: Prosa, T. J.; Moulton, J.; Heeger, A. J. *Macromolecules* 1999, 32, 4000.



Doping in organic semiconductor

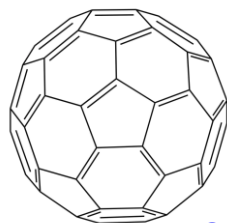
- The intrinsic charge carrier concentrations of organic semiconductor at room temperature in the dark are nearly negligible and hence the charge carrier mobility is relatively small as compared to their inorganic counterparts, they can be nearly regarded as insulator.
- However, the charge carrier concentration can be increased by “doping” yet the techniques for doping are different from inorganic materials. Unlike inorganic counterparts in which doping is accomplished by introducing of dopant *atoms* that provide addition free charge carrier at room temperature, in organic semiconductors doping is usually achieved by either the introduction of foreign *molecules* or other methods such as chemical, photochemical or electrochemical doping. These techniques introduce extrinsic charge carrier into the organic semiconductors.
- Donor or p-type materials can be doped by exposing to oxygen or other strong oxidizing agents such as iodine. The doping effect is achieved by transferring an electron from the ground state of an organic semiconductor to the oxidizing agent, resulting in an increased charge carrier concentration in the hole conductor.
- Whereas, electrochemical reducing or introducing alkali metals, phenothiazine and also semiconductors with low ionization potential are used to dope n-type materials.

Doping in organic semiconductor

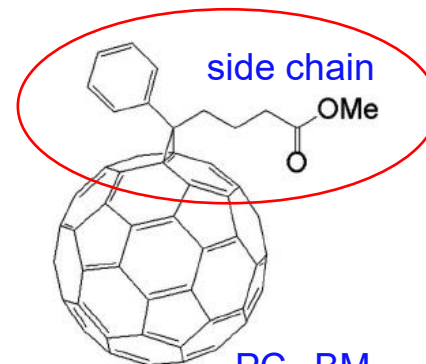
- Compared to doped inorganic semiconductors, organic semiconductors can have very few mobile charges available – particularly in the dark – so that extra mobile charges due to illumination can make a large difference. Hence, they can sometimes be described as photo-conducting insulators.
- For instance, a mixture of CuPc:C60 system, commonly used in OPV, does not show an improved behavior in charge transport properties when it is in the dark, yet it leads to an enhancement in photoconductivity under illumination. This is a result of the photo-induced charge transfer and hence this process is known as *photodoping*.

Processing of organic materials

- Organic materials processing techniques are important because it will affect the morphology of the active layer.
- There are two commonly techniques to process organic materials, i.e. **wet processing** and **dry processing**.
- Wet-processed methods include inkjet printing, screen printing, doctor blading, dip coating, spin coating etc., whereas dry processing usually refers to the thermal evaporation or sublimation.
- Small molecule organic semiconductor materials are less soluble than polymers. They can be generally sublimed in a vacuum chamber.
- The solubility of the small molecules can be improved using side-chain-solubilization, e.g., PCBM. The solubility of PCBM is improved by attaching the solubilizing groups on the bulky ball C₆₀.



C₆₀

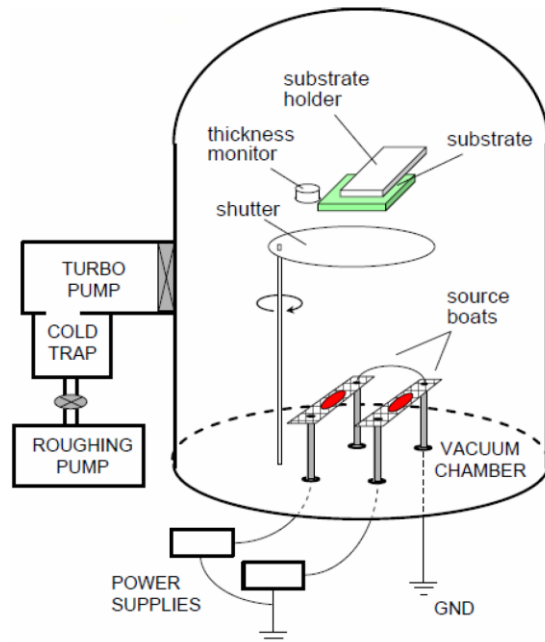


PC₆₀BM

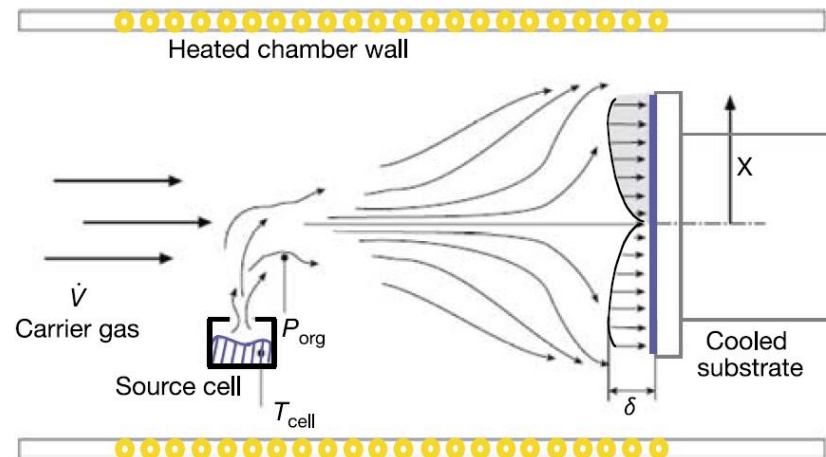
- Some short chain polymers or oligomers may also be evaporated to fabricate efficient OPV.
- On the other hand, conjugated polymers will decompose under high temperature and they are not suitable for evaporation. However, the solubility of these materials is generally good, which allows them to be processed in solution form.

Dry Processing (for small molecule materials)

- co-evaporation can be done to create an interpenetrating donor-acceptor network or to achieve molecular doping for efficient device performances
- An advantage of evaporation over wet processing is an unlimited numbers of layers can be grown in sequence.
- A drawback for dry processing is that the size of the device is limited by the vacuum chamber, affecting the potential of production upscaling.



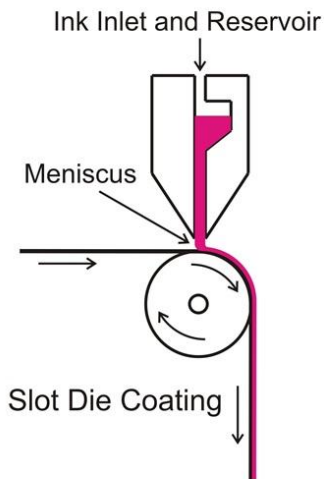
High-vacuum thermal evaporation



Organic vapour phase deposition

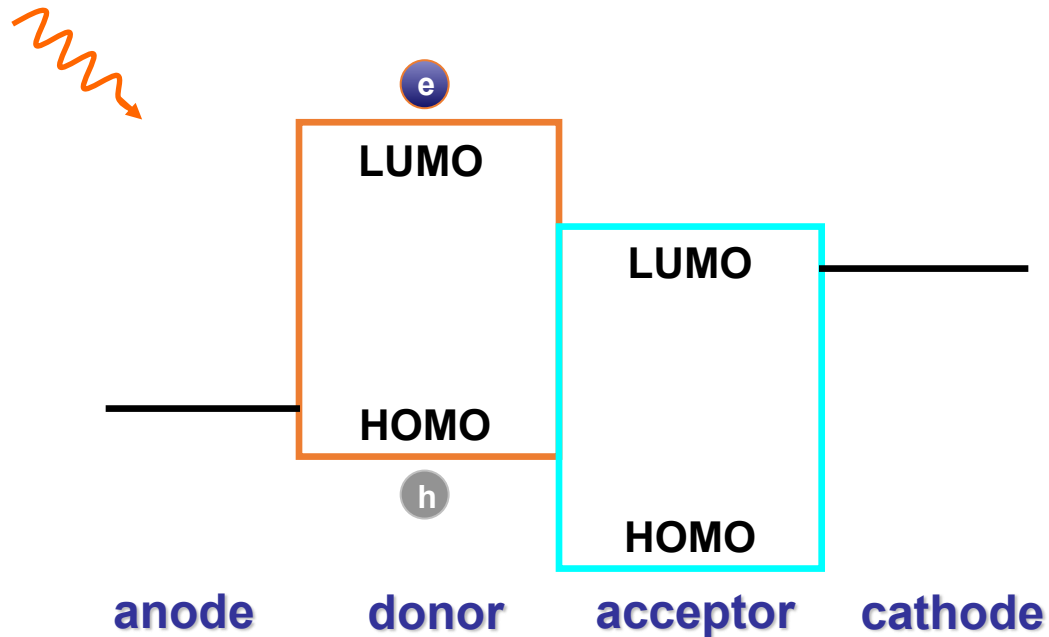
Wet processing

- Common to all wet-processed methods is the dissolution of organic semiconducting materials in an appropriate solvent such water or any other polar or non-polar organic solvents.
- After deposition, the solvent is removed by vaporization.
- Solution-processed technology is generally more preferable compared to vacuum evaporation due to the potential of production upscaling, low-cost process and low energy consumption during the manufacturing of OPV.
- The most commonly used wet processing in the laboratory is spin coating.
- However, the deposition of patterned organic film is difficult, and a considerable amount of material is flying off the substrate and wasted.
- For both film deposition and patterning, other techniques such as inkjet printing, screen printing and slot-die coating are preferred.

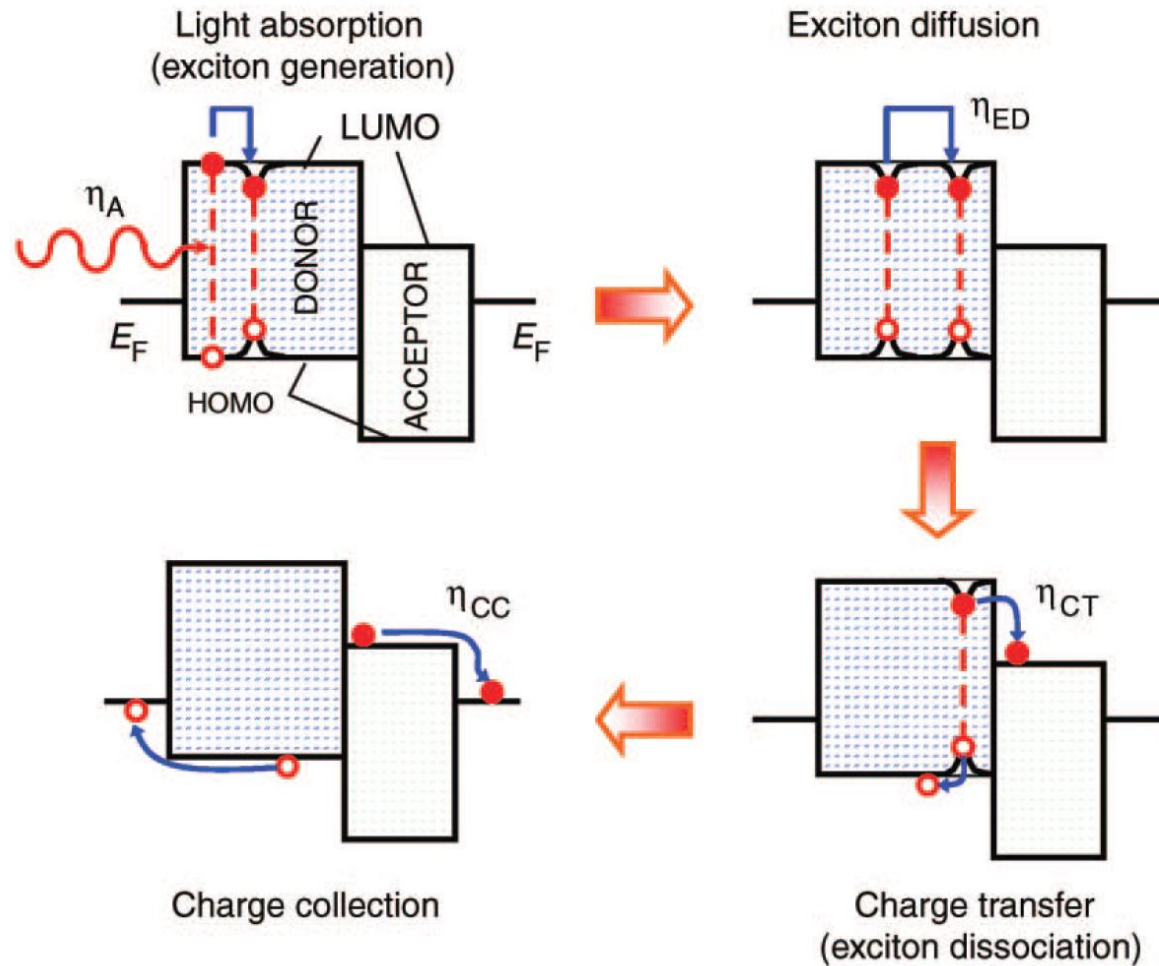


Working principle of OPV

- **Energy conversion process (five major steps)**
 - 1) Light absorption and exciton generation
 - 2) Diffusion of exciton to D-A interface
 - 3) Exciton dissociation by charge transfer at D-A interface
 - 4) Charge transport to electrodes
 - 5) Charge collection at electrodes



Working principle of OPV



Conversion Steps

Loss

1. Light absorption and
Exciton generation

Reflection, Transmission
Spectral Mismatch

η_A

2. Exciton Diffusion

Exciton recombination

η_{ED}

3. Exciton Dissociation

No charge separation and
Subsequent recombination
of excitons

~ 100% at
Appropriate
Band offset

4. Charge Transport

Recombination of charges
Limited mobility of charges

η_{CT}

5. Charge Collection

Recombination near electrodes
Energy barrier at electrodes

η_{CC}

$$\eta_{EQE} = \eta_A \times \eta_{ED} \times \eta_{CT} \times \eta_{CC}$$

$$\eta_{IQE} = \eta_{ED} \times \eta_{CT} \times \eta_{CC}$$

Light absorption and exciton generation

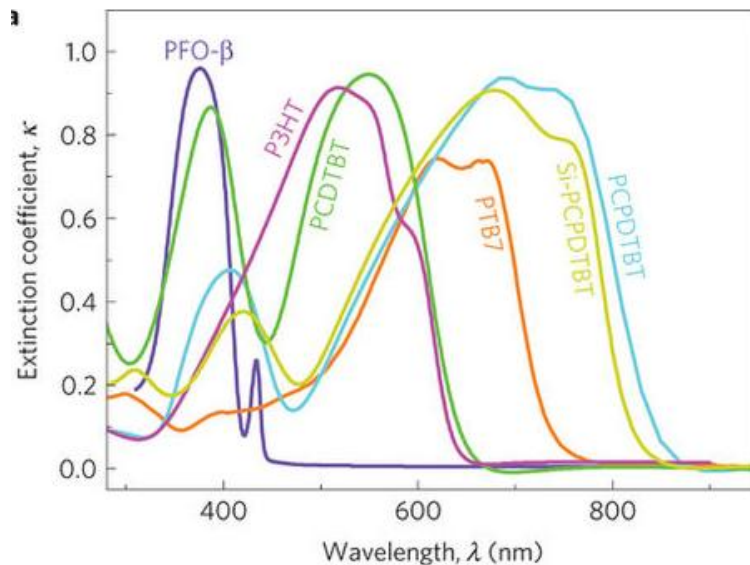
- The first step of photovoltaic conversion process in OPV is the absorption of photons with appropriate energy by organic materials leading to the generation of excitons.
- Every single photon incident on the OPV, however, does not generate exciton because of the losses. The exciton generation is generally limited by the absorption efficiency (η_A) which can be defined as

η_A = the number of photon generated excitons/ the number of incident photons

- The absorption coefficients of organic semiconductors are up to $\alpha \sim 10^5 \text{ cm}^{-1}$ in their most strongly absorptive regions, at least one order of magnitude higher than most inorganic semiconductors ($\alpha \sim 10^4 \text{ cm}^{-1}$).
- According to the equation, $\eta_A = (1 - e^{-\alpha d})$, an absorption of > 85% can be obtained in a 50nm to 100nm thick organic film in a PV cell. However, due to the mismatch factor in the refractive indices of semiconducting polymers and air, appreciable amount of the light is lost due to reflection. Moreover, 100% transmission is never achieved when incident light passes through the transparent electrode. The reflection loss at the interface of two materials and transmission loss at transparent electrode become significant in a multilayered thin film OPV because it reduces the light absorption in the active layer.

Light absorption and exciton generation

- Due to the large bandgap in most semiconducting polymers, only a small fraction of incident solar light is absorbed.
- In the case of inorganic semiconductor, e.g., Si which has bandgap of 1.1 eV is able to absorb 77% of the solar irradiation whereas most of the commercially available semiconducting polymers have bandgaps higher 2.0 eV limiting the possible absorption to only 30% of solar spectrum.
- Recently, low bandgap polymers are introduced yet their bandgaps are still higher than those of inorganic counterparts



Exciton Diffusion to D-A interface

- Once an exciton is photogenerated, it typically diffuses to a remote site. At the same time, the exciton can decay either radiatively or nonradiatively to its ground state with typical lifetimes from picoseconds to nanoseconds. Alternatively, in condensed phases, some excitons may be trapped in defect or impurity sites. As efficient exciton dissociation in the next step only occurs at the D-A interface, excitons generated far from the interface must diffuse to the interface within its life time. Hence, the performance of OPV is also limited by the exciton diffusion efficiency (η_{ED}).

η_{ED} = the number of diffusing excitons to D-A interface/ the number of photogenerated excitons

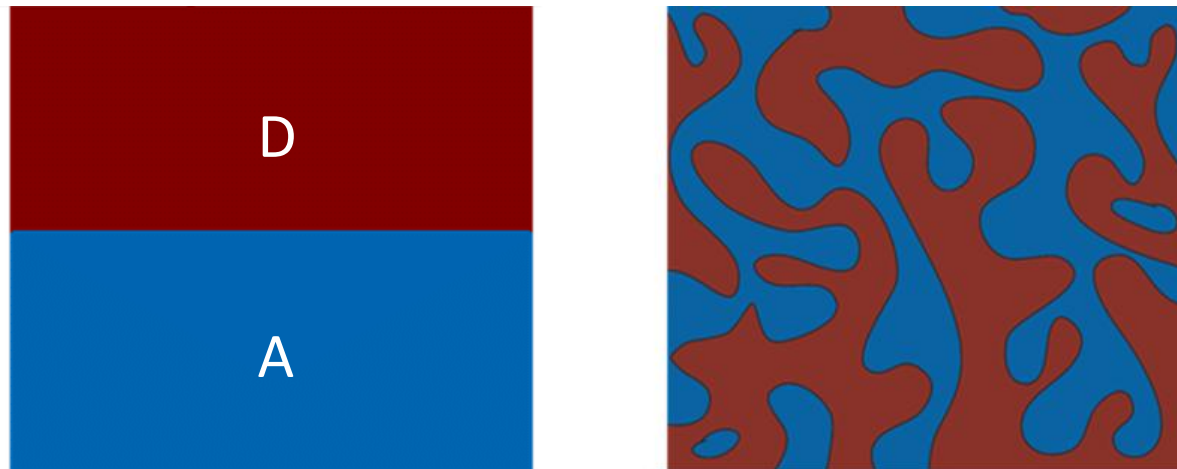
- The exciton diffusion efficiency is governed by the below equation, where d is the absorbing organic layer thickness

$$\eta_{ED} = e^{-d/L_D}$$

- The exciton diffusion length L_D , in an organic material is limited due to weak intermolecular interactions. Typically, $L_D \sim 100\text{-}200 \text{ \AA}$, is considerably shorter than the optical absorption length.

Exciton Diffusion to D-A interface

- Therefore, although in an organic D-A heterojunction that is thick enough to absorb a significant portion of the incident light, only a small fraction of photogenerated excitons can reach the D-A interface, leading to $\eta_{ED} \ll 100\%$.
- This efficiency can be greatly enhanced in a spatially distributed D-A interface that intermingles the donor and acceptor molecules where the **D-A phase separation length is the same order of magnitude as the exciton diffusion length**.

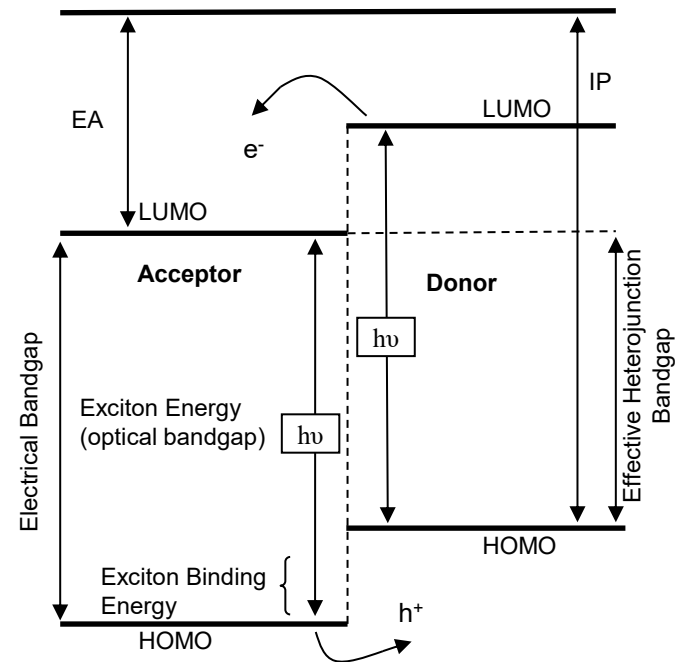


Exciton Dissociation

- As the electron and hole in the exciton are not free charge carriers, a driving force must exist to dissociate it to free electrons and holes. A strong electric field is able to overcome the binding energy of the exciton and assist the exciton dissociation. However, with a typical built-in electric field in the organic PV cells, a field-induced exciton dissociation efficiency of $\eta_{\text{FIED}} < 10\%$ is estimated. Therefore, other driving forces need to be employed for efficient exciton dissociation.
- It has been observed that without an electric field, efficient exciton dissociation can also occur at certain organic semiconductor heterojunction with appropriate **energy level offset**. In such heterojunctions, exciton induces electron transfer to the larger *electron affinity* (χ) material and holes transfer to the smaller *ionization potential* (IP) materials. In other words, larger electron affinity material and smaller ionization potential material serve as 'electron acceptor' and 'electron donor' respectively, and the interface is commonly referred as organic donor-acceptor (D-A) interface. **The dissociation of the strongly bound exciton is energetically favorable when the HOMO and LUMO level offsets between the donor and acceptor materials are sufficiently large to overcome the exciton binding energy.** Through the charge-transfer process, a hole in the HOMO of the donor and an electron in the LUMO of the acceptor molecule are created by the dissociation of an exciton. As the driving force only exists where the two materials are in close contact, only excitons within the exciton diffusion length can be effectively dissociated.

Exciton Dissociation

- It is reported that **fullerene and its derivatives are strong electron acceptors**. They are able to break photoexcited excitons apart into charge carriers efficiently.
- Ultrafast photophysical studies showed that the photoinduced charge transfer in such blends happens within a time scale of 45 fs. This is much faster than other relaxation process such as photoluminescence (usually occurs at around 1 ns). The reverse electron transfer from fullerenes back to the polymer is extremely slow. Thus, **the quantum efficiency of this dissociation process is close to 100% at the polymer-fullerene interface**.



Charge transport towards electrodes

- Once the excitons are successfully dissociated, the charge carriers need to transport towards the electrodes.
- These charge carriers are unable to travel independently unless a driving force is applied. The driving force may include the electrical field created by a chemical potential gradient and the work function difference between the two electrodes.
- A chemical potential gradient in electrons and holes is built up at the D-A interface. This gradient is generally defined by the difference between the HOMO of donor and LUMO of acceptor. This internal electric field determines the maximum V_{oc} and contributes to a field-induced drift of the charge carriers.
- Even without built-in potential, i.e, two electrodes are the same material; the chemical potential driving force can also induce the asymmetric photovoltage. However, in order to enhance the charge transport, asymmetrical contacts are used, e.g., a cathode with a low work function for electron collection and an anode with a high work function for hole collection, to generate an external field in short circuit conditions according to a metal-insulator-metal configuration.

Charge transport towards electrodes

- The charge carriers may recombine with a rate related to the device structure and the morphology of the active layer. The donor and acceptor have to generate a percolation pathway for photogenerated charge carriers to travel before they are successfully collected by the electrodes.
- In a planar D-A heterojunction, the photogenerated holes and electrons are spatially separated, so that the recombination between them is unlikely. Hence, the charge transfer efficiency η_{CT} is near unity.
- For bulk heterojunction, the photogenerated electrons and holes have a significant spatial overlap. With lower charge carrier mobilities in the mixed layer than in homogeneous layers, recombination of charges in the mixed layer can be very significant. Hence, $\eta_{CT} < 1$.
- η_{CT} = the number of dissociated excitons transported to electrodes / the number of dissociated excitons at D-A interface

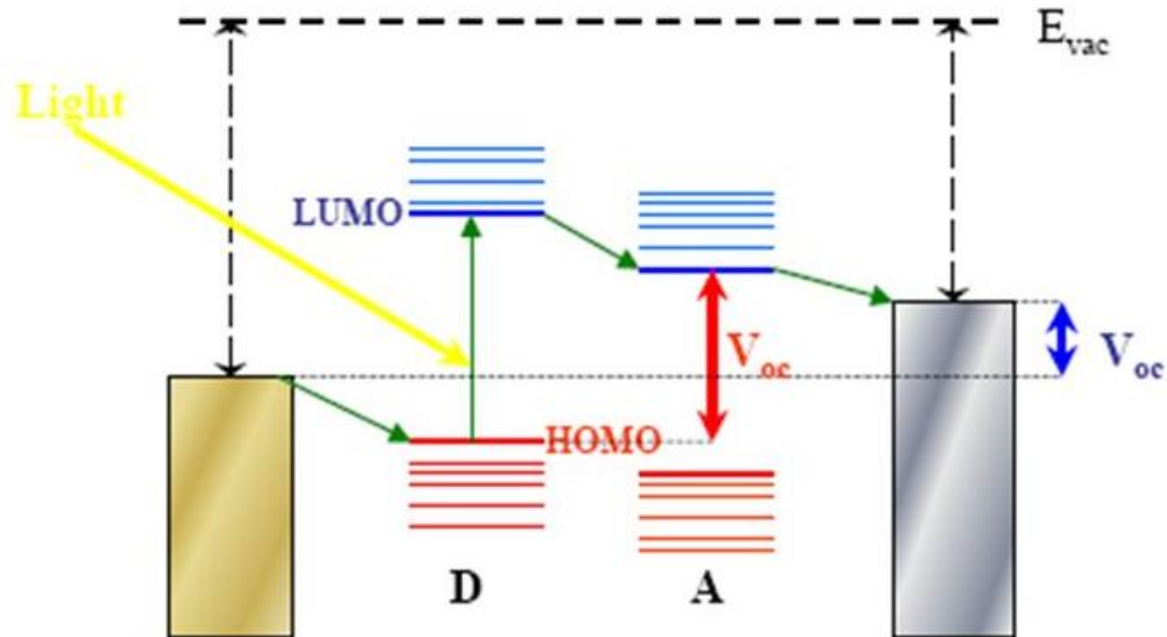
Charge collection at the electrodes

- In this last step of the photocurrent generation, the free electron and holes reaching the electrodes will be collected by the electrodes and flow to the external loading circuit.
- When the acceptor LUMO level matches the Fermi Level of the smaller work function electrode (Cathode), and the donor HOMO matches the Fermi level of the larger work function electrode, an ideal ohmic contact would be established for efficient carrier collection at the electrodes.
- ITO is commonly used as a substrate for both OLEDs and OPV as the high work function electrode. It is a highly degenerate n-type semiconductor with low resistance, high work function and high transmittance in the visible.
- Evaporated metals such as silver (Ag) or aluminum (Al) are used as cathode.
- So far, there are no OPV cells that have achieved perfect ohmic alignment due to the availability and limitations of materials and electrodes involved.
- The interfacial band bending, energy offset and chemical effect at the metal-organic interface also affects the carrier collection.

Photovoltage in OPV

- Light harvesting materials employed in OPV have an optical band gap of around 1.7 to 2.1 eV and yet the V_{OC} barely exceeds 1.0 V, which is approximately just half of the photon's original energy.
- By contrast, in inorganic counterparts such as Si, CIGS and GaAs, the difference is only 0.3 to 0.45 eV between the material bandgap and V_{OC} .
- Major V_{OC} loss comes from the band offset between donor and acceptor, which is required for charge dissociation.
- Therefore, the energy level difference between the LUMO of acceptor and the HOMO of donor ($E_{LUMO,A} - E_{HOMO,D}$), which is known as the effective bandgap, determines the V_{OC} of OPV, rather than the optical band gap of light absorber.
- On the other hand, the V_{OC} is also determined by **the work function difference between two electrodes**, according to the metal-insulator-metal (MIM) model.

Photovoltage in OPV



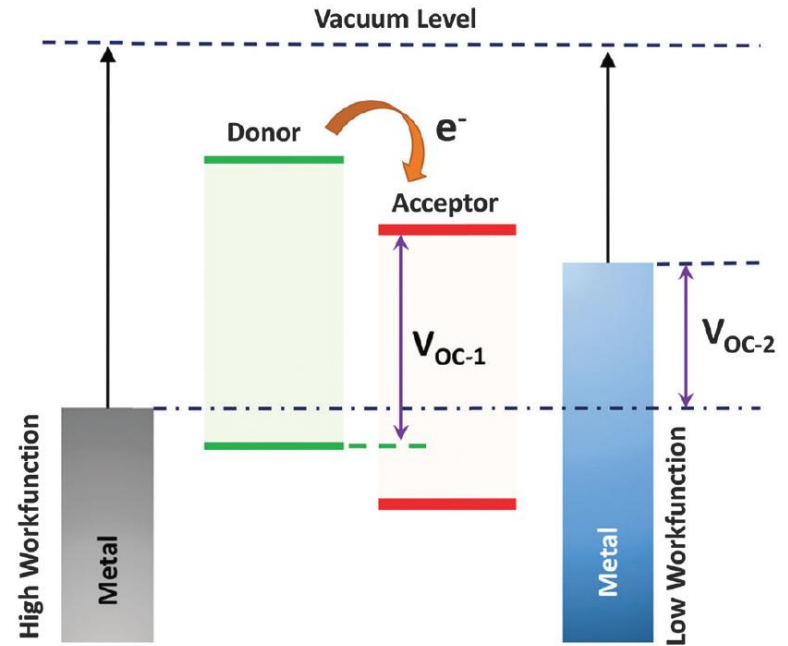
Which is the V_{oc} ?

Is it in the **electrodes**?

Is it in the **donor/acceptor**?

Photovoltage in OPV

- When the contacts are ohmic (the Fermi level of contact lies close to either $E_{\text{LUMO,A}}$ or $E_{\text{HOMO,D}}$), V_{OC} is governed by the Donor–Acceptor interface ($V_{\text{OC-1}}$).
- This is because for ohmic contact, the Fermi level of the cathode (anode) is **levelled and pinned** to the $E_{\text{LUMO,A}}$ ($E_{\text{HOMO,D}}$)
- When the contacts are non-ohmic, V_{OC} is determined by the work function difference of the electrode following the metal–insulator–metal (MIM) model ($V_{\text{OC-2}}$).



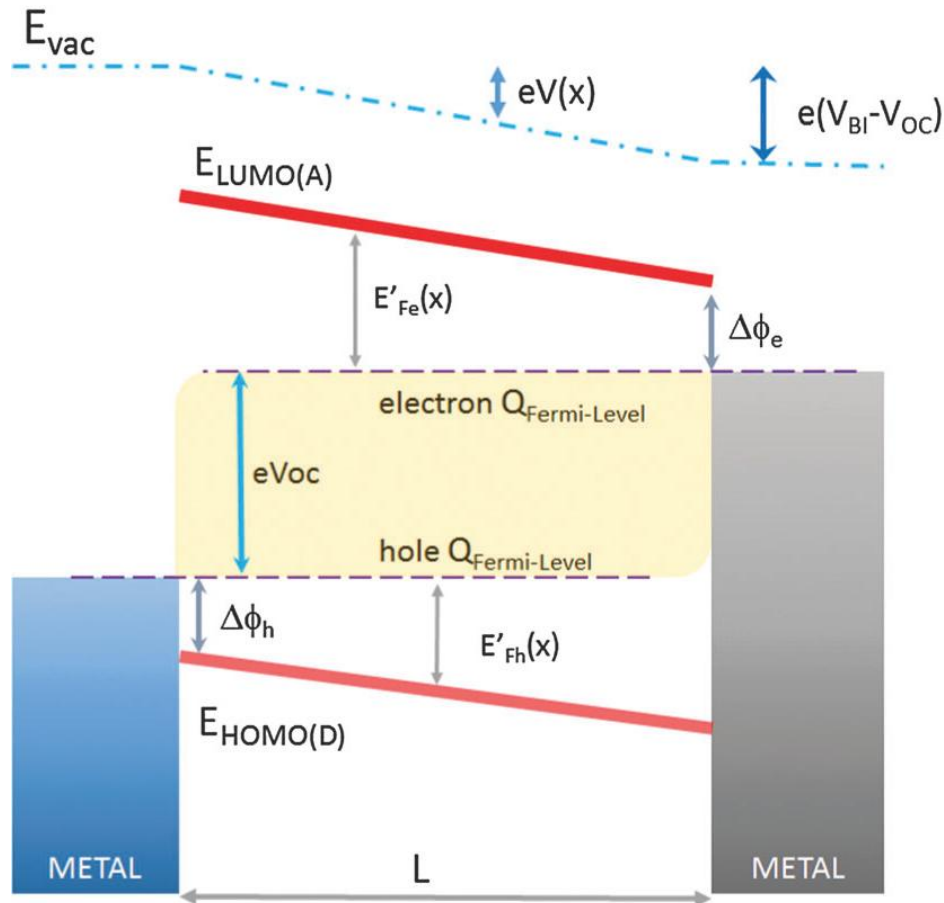
Photovoltage in OPV

- In general, it can be said that the V_{OC} of any kind of solar cell originates from the splitting of electron and hole quasi-Fermi levels actuated by light illumination.

$$V_{OC} = 1/q (E_{Fn} - E_{Fp})$$

- However, unlike crystalline inorganic semiconductors, which exhibit well-conceived hole and electron quasi Fermi-levels, the organic semiconductors possess disorder induced gap tail states. Therefore, the distribution of photogenerated carriers in such gap tail states downshifts the electron quasi-Fermi level and upshifts the hole quasi-Fermi level, thereby reducing V_{OC} correspondingly.
- Hence, direct determination of electron and hole quasi-Fermi levels in OPV is quite elusive.
- Moreover, there are many other factors influence the V_{OC} of OPV; density of states or energetic disorder, charge transfer states, donor-acceptor interface, microstructure, carrier density etc.

Photovoltage in OPV



Energy-level diagram of an OPV device at the open-circuit voltage. E_{vac} is the vacuum level. $\Delta\phi_h$ and $\Delta\phi_e$ are the hole and electron injection barriers, respectively. V_{BI} is the difference in the contact work functions. $E'_{Fh}(x)$ and $E'_{Fe}(x)$ are hole and electron quasi-Fermi potentials relative to their respective energy levels.

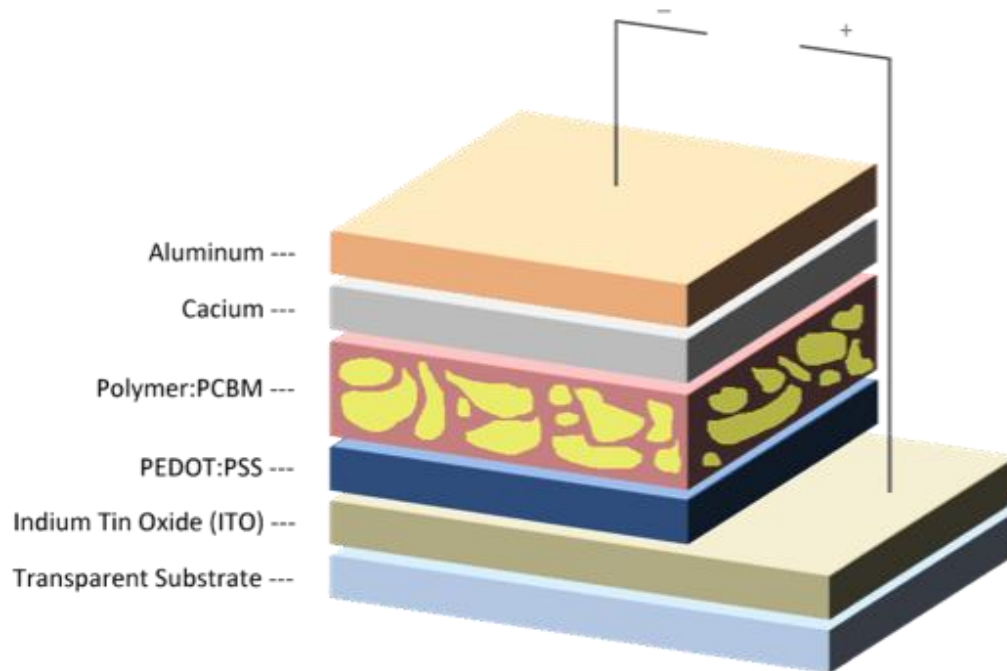
Photovoltage in OPV

- The empirical relationship between the HOMO of donor and the LUMO of acceptor was proposed by

$$V_{OC} = 1/q (E_{LUMO,A} - E_{HOMO,D} - \Delta)$$

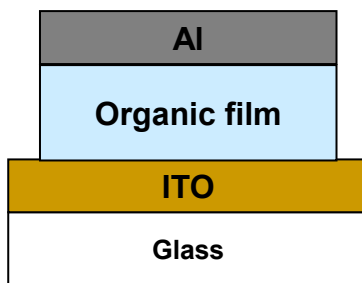
- The empirical value of Δ based on a wide range of donor and acceptor combination is 0.3 V but it can be greater or smaller than this, depending on the influential factors.
- $V_{oc} = \frac{nkT}{q} \ln \left(\frac{I_{ph}}{I_0} + 1 \right)$ is still valid for OPV. Hence, the slope of V_{OC} vs. light intensity in logarithmic scale gives the slope of nkT/q , with n is close to 1 for bimolecular recombination while $n > 1.5$ when other recombination mechanism such as trap-assisted recombination.

Evolution in Device Architecture

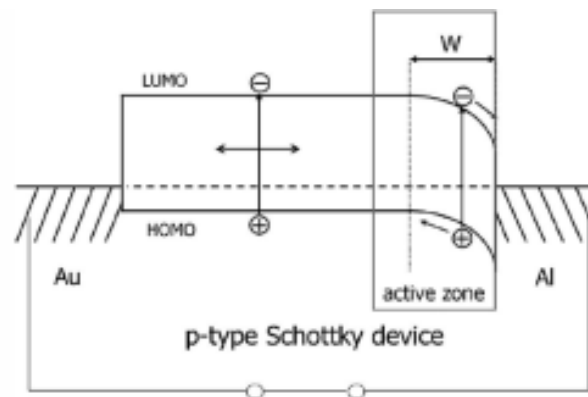


Single Layer Device

- In its simplest form, an OPV consists of a single polymer layer sandwiched between two different electrode materials.
- The rectifying behavior of the device can be explained by [formation of Schottky barrier](#) between the metal with the lower work function and the p-type organic layer.
- In the Fig., the energy band diagram is illustrated for the case of a Schottky junction at the aluminum contact. Close to the contact, in the depletion region W , energy band is bent due to an electric field in which excitons can be dissociated.
- Because the exciton diffusion length for most OPV materials is below 20 nm, only those excitons generated in a small region within <20 nm from the contacts contribute to the photocurrent. Due to high series resistances, these materials show a low FF and a field-dependent charge carrier collection.

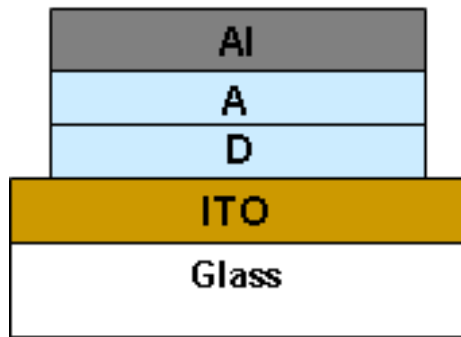


(a)

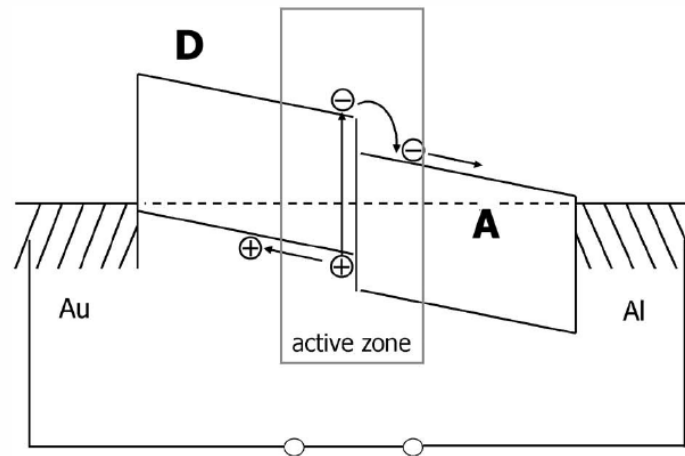


Bilayer Device

- In a bilayer heterojunction device, the donor and acceptor layers can be prepared sequentially by thermal evaporation or by solution casting one material followed by evaporation of another material or by sequential solution casting (solvent with orthogonal solubility to the underlying layer must be used). PCE of about 4.2% under AM1.5G illumination was reported for OPV with a thermally evaporated bilayer device using CuPc and C60.
- In a bilayer device, a donor and an acceptor material are stacked together with a **planar interface where the charge separation occurs**. This process is mediated by a large potential drop between donor and acceptor.



(a)

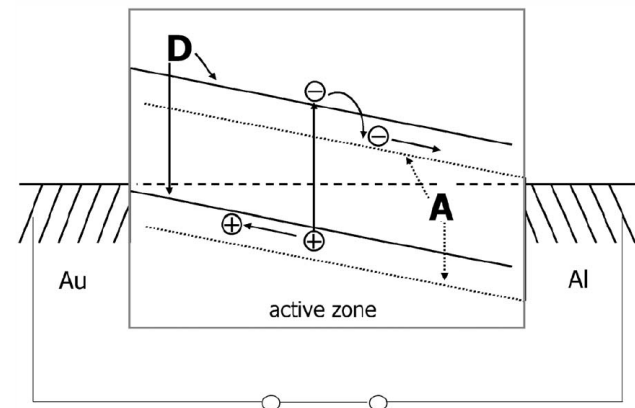
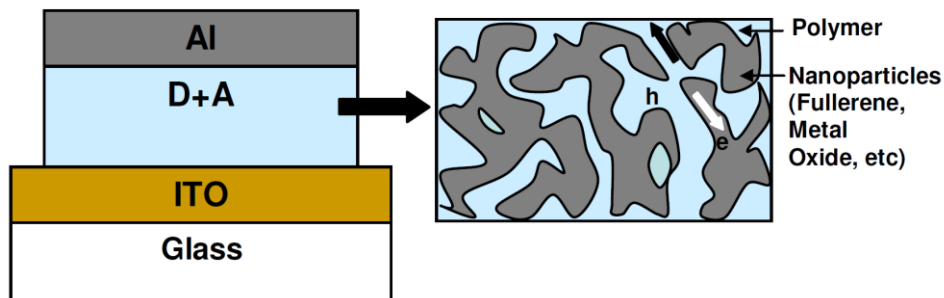


Bilayer Device

- The bilayer is sandwiched between two electrodes matching the donor HOMO and the acceptor LUMO, for efficient extraction of the corresponding charge carriers.
- The advantage of bilayer device is the monomolecular transport of the charge carriers after the dissociation of the excitons. The electrons travel within the n-type acceptor and the holes travel within the p-type donor material. Hence, the opposite migration of charge carriers will prevent the holes and electrons from recombination. As a consequence, the photocurrent dependency on illumination intensity can be linear, and for thinner layers, larger filling factors can be achieved.
- In bilayer donor-acceptor PV cells, a total thickness L is typically of the order of optical absorption length L_A , therefore $\eta_A = 1 - \exp(-L/L_A) > 50\%$ if optical interference effects are ignored.
- And η_{CT} is close to 100% due to the relative high donor and acceptor mobility in their respective neat layers and the small recombination rate.
- However, since the exciton diffusion length (L_D) is typically an order of magnitude smaller than L_A , a large fraction of the photogenerated excitons cannot reach the dissociation site. Therefore they are recombined and become unused for photocurrent generation, limiting the PCE of the planar junction cell.

Bulk-heterojunction Device

- The limitation of exciton diffusion to D-A interface in bilayer device was finally overcome by the introduction of bulk heterojunction (BHJ) concept.
- In a BHJ device, the donor and acceptor are blended together in a bulk volume. By proper device processing control, the blend exhibits a donor-acceptor phase separation within a distance less than the exciton diffusion length of each absorbing site. Hence, **the excitons are efficiently dissociated at the interface within its lifetime** and recombination is reduced to a great extent, leading to higher photocurrent density.



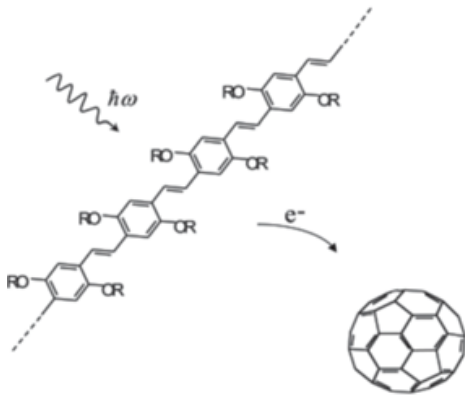
Bulk-heterojunction Device

- Another advantage of the BHJ concept is that it exhibits **a large interfacial area where exciton dissociation can occur**. Unlike bilayer device in which active zone is only within an exciton diffusion length, in BHJ device the active zone extends throughout the full absorption region and charge separation can take place throughout the whole photoactive layer.
- The BHJ device, however, requires percolation pathways for the holes and electrons to be collected at the respective electrodes. In other words, the donor and acceptor phase have to form a **bicontinuous and interpenetrating network**. Therefore, as a main limitation, a BHJ device is more sensitive to the nanoscale morphology than a bilayer device.
- Generally, BHJ may be achieved by co-evaporation of donor and acceptor small molecules, or solution casting of acceptor-donor blend, either polymer/polymer, molecule/molecule or polymer/molecule.

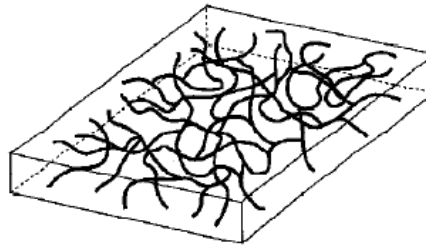
Bulk-heterojunction Device

Why BHJ concept works?

- Ultra-fast charge transfer between donor and acceptor (conjugated polymer and fullerene)
- Self-assembly of donor and acceptor materials by spontaneous phase separation

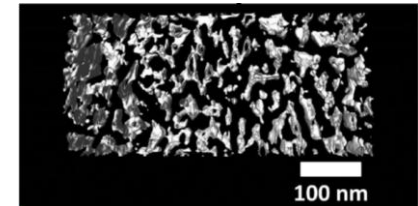


Concept (1995)



G. Yu et al, Science, 270, 1789 (1995)

Reality (Now)

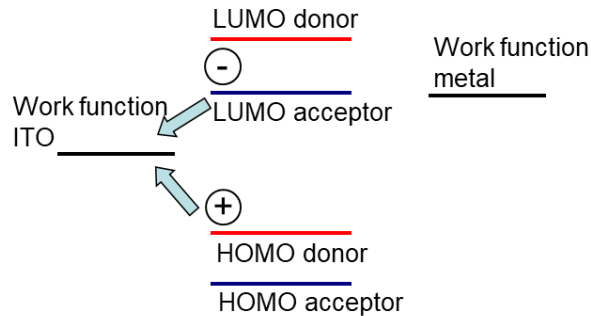


A. J. Heeger, Adv Mat., 26, 10 (2014)

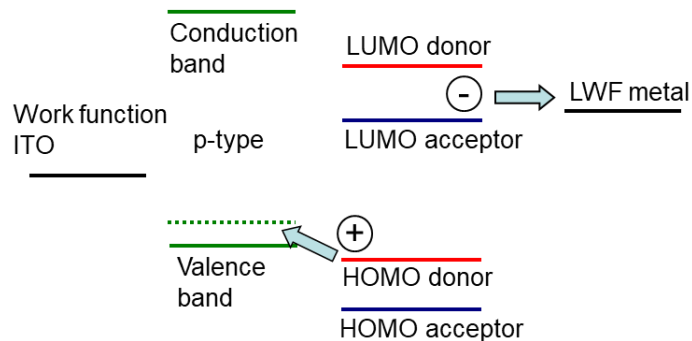
	Bulk-heterojunction	Bilayer Heterojunction
Advantage	Charge dissociation is very efficient due to large interfacial layer The whole thickness of the film can generate photocurrent	Less recombination (electrons and holes spatially separated) Efficient charge transport and collection
Disadvantage	Bi-continuous and interpenetrating network of donor and acceptor phases is required for efficient charge transport More recombination (electrons and holes have significant spatial overlap)	Charge dissociation is not efficient Only the region in a diffusion length away from the interface can generate photocurrent

BHJ Device (conventional vs inverted structure)

(a) no buffer layer

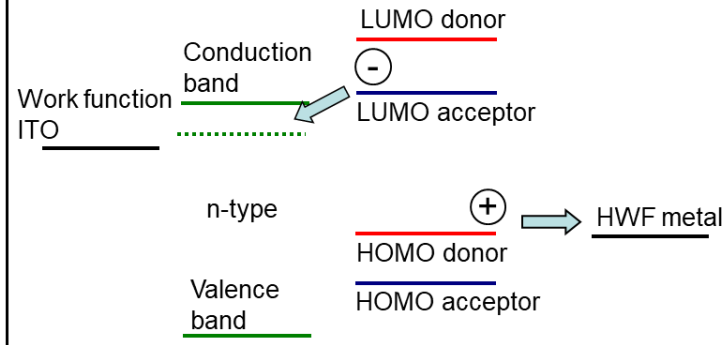


(b) with p-type buffer layer



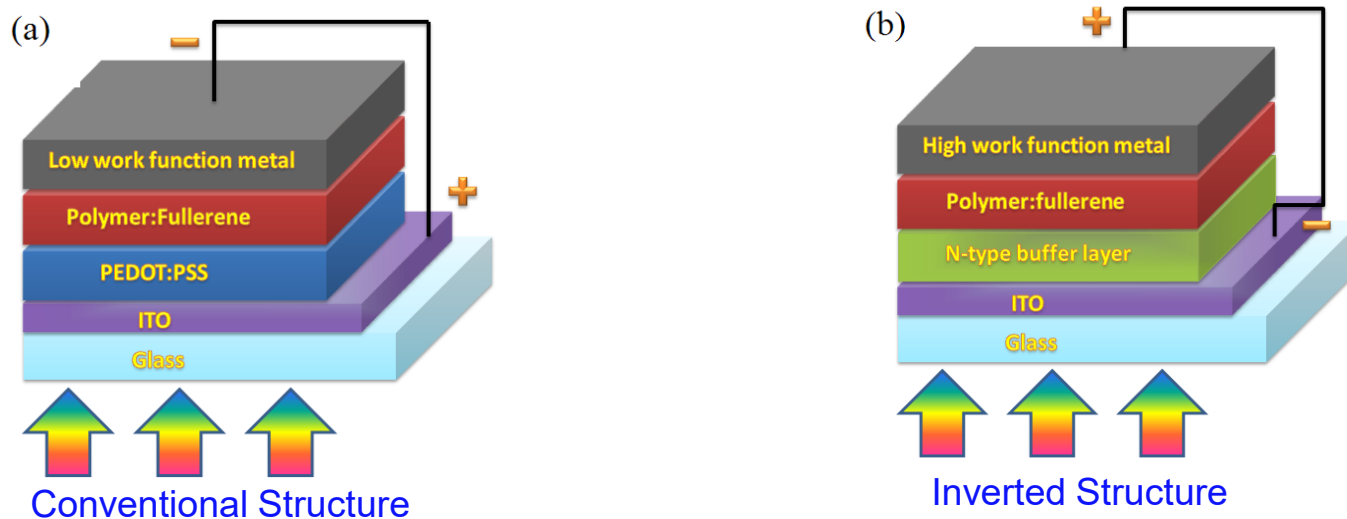
Conventional Structure

(c) with n-type buffer layer



Inverted Structure

Conventional vs Inverted structure



- The energy conversion process of incident photons into electricity in an inverted structure device is the same as that in a conventional.
- Only difference between two structures is their nature of charge collection.
- Typically, a conventional structure device consists of ITO/PEDOT:PSS/polymer:fullerene blend (BHJ)/LWF metal from bottom to top (Note: in some device structure, cathode buffer layer (Ca, LiF etc.) are inserted between the BHJ and LWF metal.)
- An inverted structure device is composed of ITO/n-type material/polymer:fullerene blend/HWF metal

Conventional vs Inverted structure

- In principle, ITO is able to collect either holes or electrons from polymer:fullerene network because its work function (from - 4.5 to - 4.9 eV) is between the HOMO and LUMO values of common organic materials
- In the conventional device, ITO is modified by p-type PEDOT:PSS, which has a HWF so that electrons are blocked and only holes are collected by ITO anode
- Since the electrons must be extracted from the top electrode, air-sensitive LWF metal is applicable as the top cathode.
- On the other hand, if ITO is modified by n-type material with a LWF, the holes will be blocked and electrons will be exclusively collected by ITO electrode.
- Collecting the holes at the other side allows the usage of HWF metal as top electrode without losing asymmetric electrode structure, eliminating the PEDOT/PSS interface issue concurrently.
- Applying this concept, in an inverted device, ITO is coated with n-type material which can be either transition metal oxides, or alkali-metal compound, or ultra thin LWF metal, or organic material.
- In some modified inverted devices, p-type-like layer or hole-transporting layer is inserted between active layer and HWF electrode to effectively block the excitons and adjust optical field distribution in the active layer

Conventional vs Inverted structure

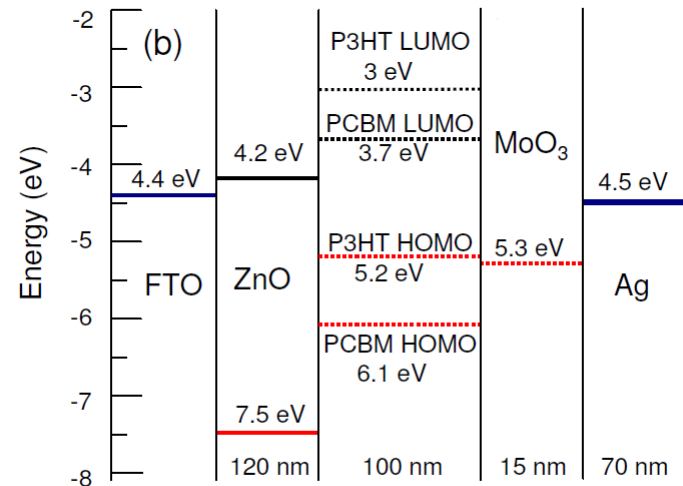
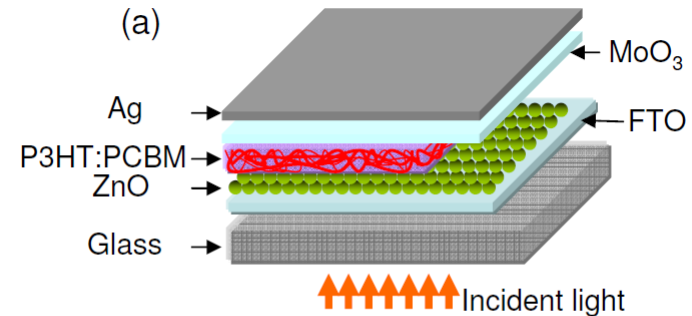
Advantages of Inverted structure

- Stability in air
 - ☐ Al is used as metal electrode in conventional device. Al is easy to oxidize because of the low workfunction nature of electrode. Moreover, diffusion of oxygen into the BHJ through pinholes and grain boundaries in Al cathode causes the degradation of BHJ layer.
 - ☐ In inverted structure, high work function metal such as Ag is used. Hence device stability is improved.
- Degradation between ITO/PEDOT:PSS interface
 - ☐ PEDOT:PSS usually destroy the underlying ITO in long-term due to the acidic nature.
 - ☐ ITO/PEDOT:PSS interface is eliminated in inverted structure
- Improved charge collection
 - ☐ Charge collection is also enhanced in the inverted geometry owing to inherent vertical phase separation with a donor-enriched top surface and an acceptor-enriched bottom surface in the blend polymer.

Inverted structure

Example of an inverted structure

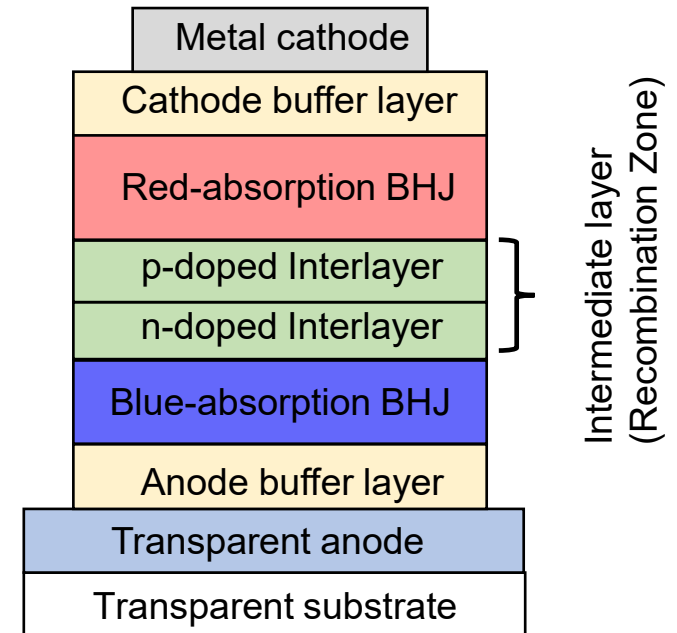
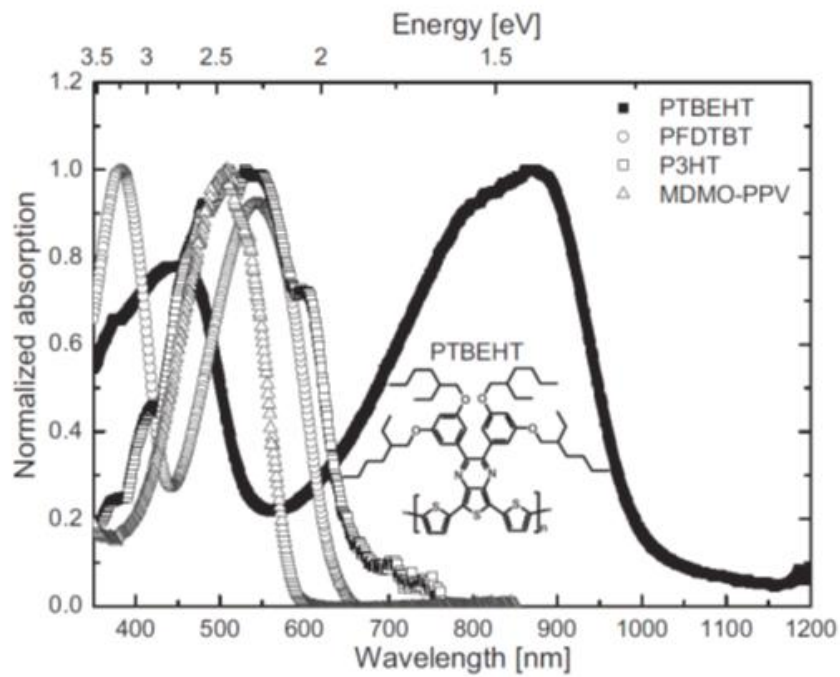
- Sol-gel processed ZnO is used as electron transport layer
- As ZnO is n-type material with CB of 4.2 eV, it transports the electron to the electrode via its conduction band.
- ZnO also effectively blocks the hole due to its large band gap (> 3 eV)
- MoO_3 is used as hole transport and exciton blocking layer.
- Strictly speaking, MoO_3 is n-type material but its CB is so deep that hole can be transported through its CB. Thus, it is called **p-type like material**.



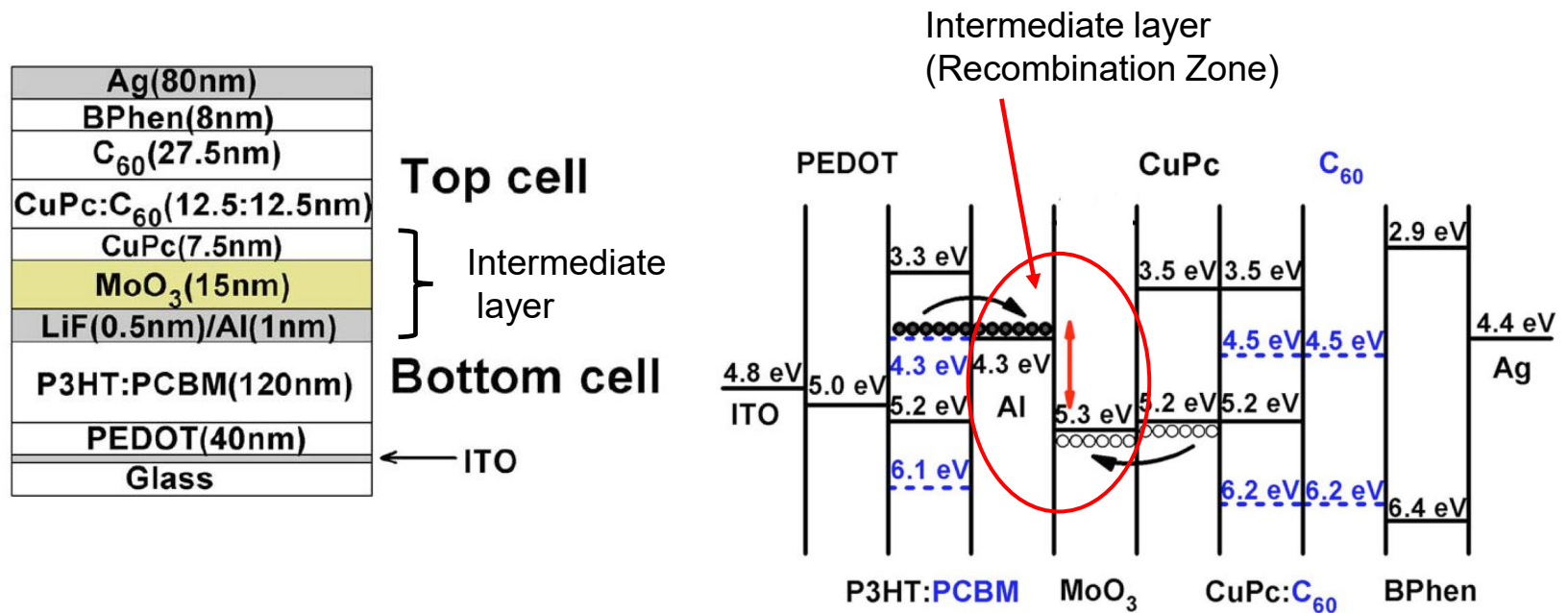
Tandem structure

Why tandem structure in OPV?

- Narrow absorption range of organic semiconductor
- Short exciton diffusion length and low carrier mobility limit the thickness of the light absorption layer



- The monolithically integrated tandem cell is composed of two (or more) devices (with complementary absorption spectrum) stacked on top of each other.
- The two cells are connected by an **intermediate layer**, which is employed to allow recombination of the holes coming from the one cell with electrons coming from the other.
- It is important that the intermediate layer should ensure the alignment of the charge carrier transport energy levels of both sub-cells and hence lossless recombination of electrons and holes.



- Since there is no voltage loss across this intermediate layer, the V_{oc} of the tandem cell in a serial connection should be the summation of the V_{oc} 's of the individual cells.

$$V_{oc,tandem} = V_{oc,1} + V_{oc,2} + V_{oc,3} \dots$$

- However the current is limited by the lower current among the sub-cells. Hence, thickness tuning is required to match the current between the top cell and bottom cell.
- The intermediate layer should possess the following properties: (1) low electrical resistance; (2) high optical transparency in the visible and infrared ranges; (3) low barriers for both electron and hole extractions; (4) easy-fabrication process; and (5) protection for the prior-deposited active layer in solution-processed tandem cell.
- The record high efficiency of OPV (in 2013) with PCE of 12% was achieved by vacuum-processed tandem OPV fabricated by Heliatek.

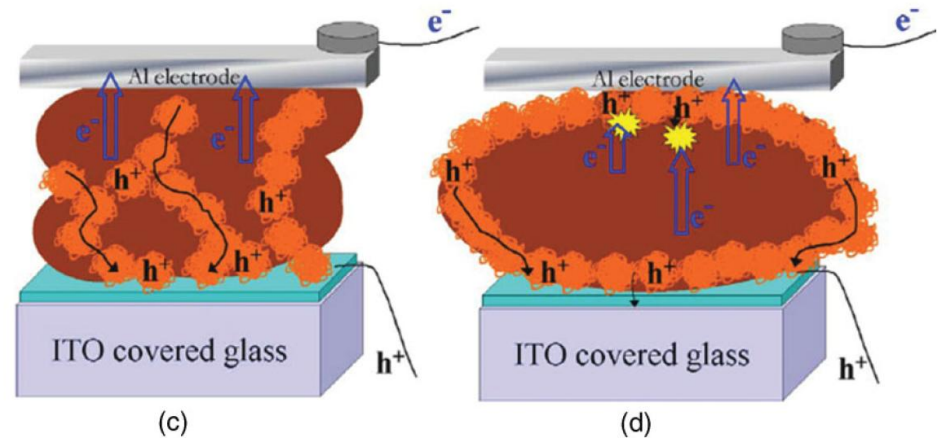
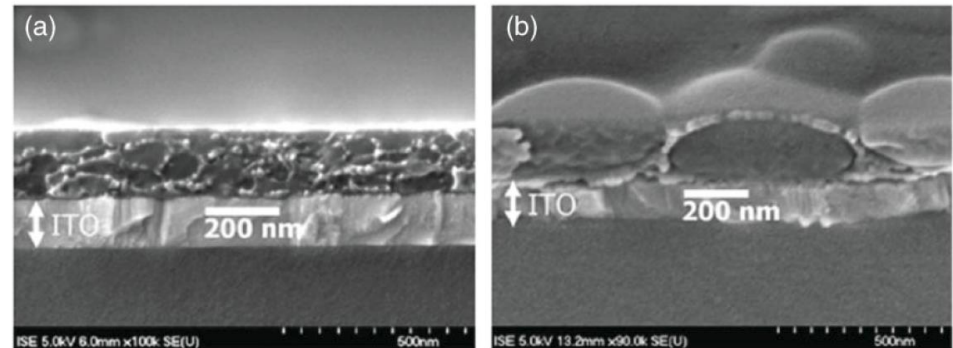
Further reading for Tandem OPV:

- L. Zang (ed.), Energy Efficiency and Renewable Energy Through Nanotechnology, Green Energy and Technology, DOI: 10.1007/978-0-85729-638-2_3, Chapter 3 “Organic solar cells with inverted and tandem structure”
- Ameri T, Dennler G, Lungenschmied C et al (2009) “Organic tandem solar cells: a review” Energy Environ Sci 2:347–363
- Hadipour A, de Boer B, Blom PWM (2008) “Organic tandem and multi-junction solar cells” Adv Funct Mater 18:169–181

Morphology control in BHJ

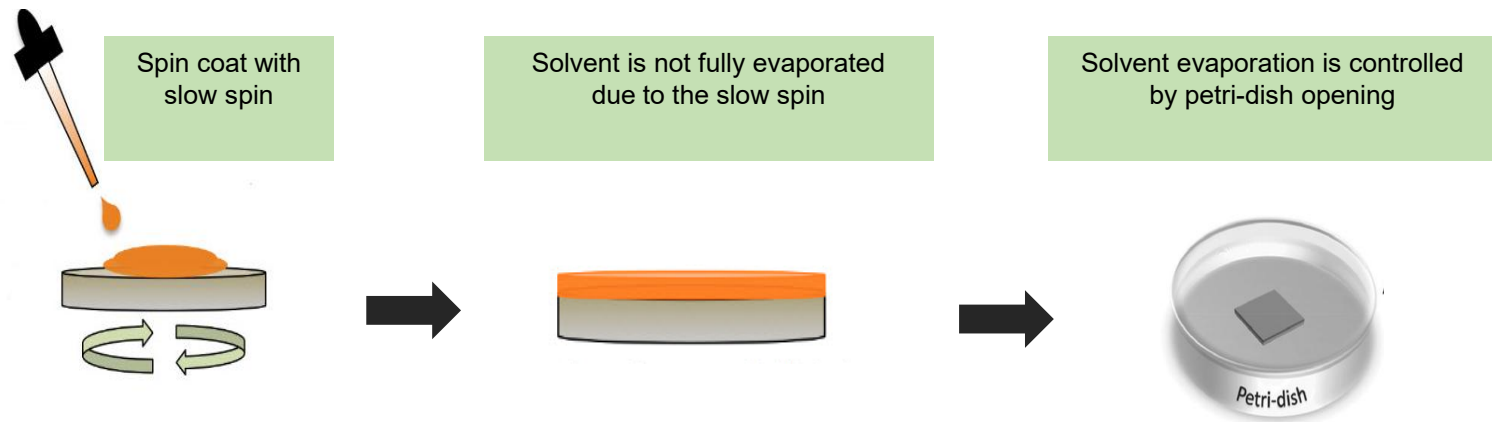
Effect of solvent

- Solvent properties such as boiling point, vapour pressure, solubility and polarity, have considerable impact on the film morphology.
- The solvent with higher solubility results in a finer phase separation.
- For example: MDMO-PPV:PCBM system, the solubility of polymer blend is much better in chlorobenzene (CB) than in toluene. Thus, CB yields a more intimate mixing and stronger interchain interaction than toluene does, which accounts for higher efficiency.
- A thick “skin” layer observed in the toluene-casted film, in which the PCBM nanocrystallites were generally covered by this “skin” layer, identified as polymer nanospheres. (b & d)
- However, for most chlorobenzene-cast films, the polymer nanospheres were homogeneously distributed. (a & c)



Effect of annealing

- Two types of annealing in OPV: thermal annealing and solvent annealing
- **Thermal annealing** can be applied either on the final device (post annealing) or on the polymer film only (pre-annealing). The annealing temperature and time are the two most critical parameters in this approach. Thermal annealing usually improve the crystallinity of polymer.
- **Solvent-annealing** approach controls the polymer nanomorphology through controlling the solvent evaporation rate. In general, a slow- evaporation rate makes the crystallization process slower and hence, improves the molecular ordering of the polymer chain.



- Various annealing processes can dramatically improve the crystallinity, resulting in higher hole mobility;

Effect of Additives

- For some polymers, annealing is incapable of improving the device performance or thermal annealing is not desirable.
- In such cases, mixing two or more miscible solvents or the incorporation of additives into a host solvent is alternative choice to control the morphology
- Examples of additives: alkanedithiols, 1-chloronaphthalene (CN), 1,8-diiodooctane (DIO)
- Two criteria for incorporating additive to control the morphology: (1) selective solubility of the fullerene component, and (2) a higher boiling point (lower vapour pressure) than the host solvent.
- Examples of control morphology with additives:
- **Example (1)**, PCPDTBT:fullerene dissolved in CB with 1,8-octanedithiol (OT). In this system, the OT selectively dissolved the fullerene phase, while the PCPDTBT was relatively insoluble. Due to the higher boiling points of the OT (b.p. $>160^{\circ}\text{C}$), the fullerene phase stayed in the solution longer than the polymer, providing more freedom to self-align and crystallize.

- Example (2)**, P3HT:PCBM dissolved in dichlorobenzene (DCB) with OT: In this system, The host solvent DCB has a lower boiling point (198 °C) than OT (270 °C), but a higher solubility for PCBM. As a result, the OT concentration gradually increased during the spin-coating, with PCBM forming clusters and aggregates in the OT phase simultaneously. P3HT has a higher surface energy than PCBM. Thus, in order to reduce the overall energy, P3HT tends to accumulate at the top (air) surface, while PCBM correspondingly segregates at the bottom. This causes the vertical phase-separated morphology with P3HT-rich region near the surface and PCBM-rich region at the bottom.

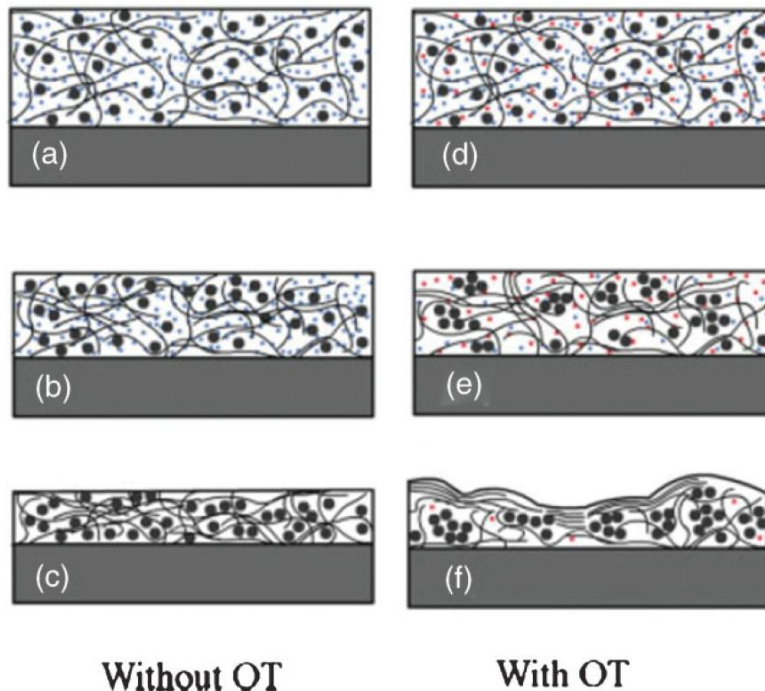
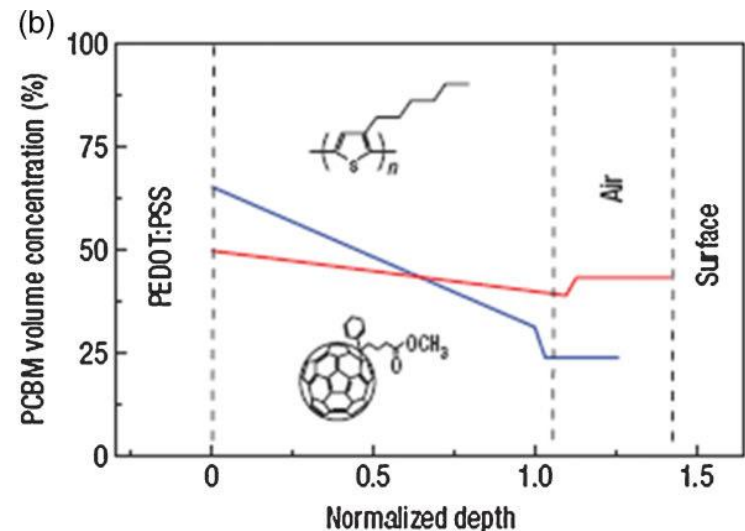
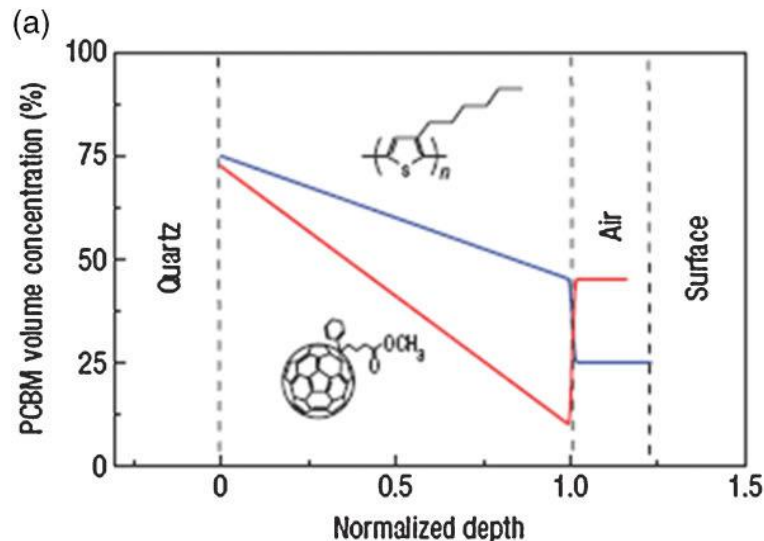
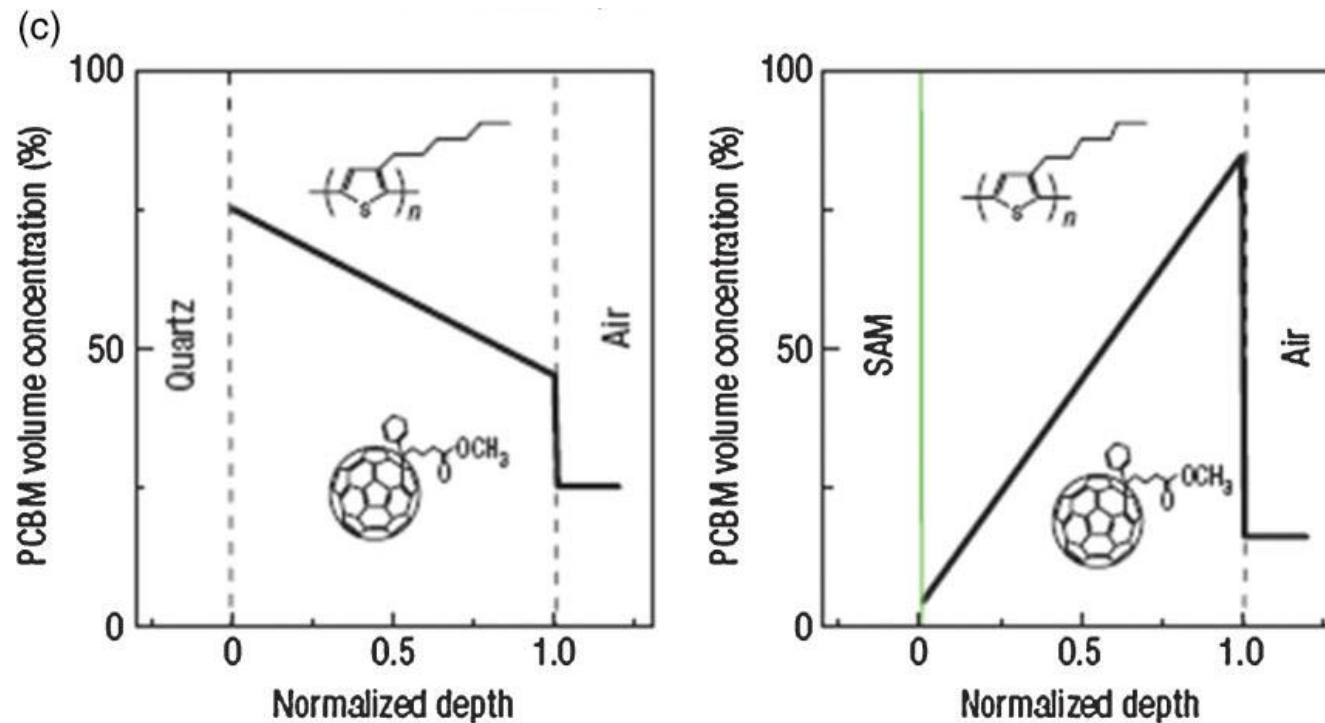


Figure: Proposed model of film evolution during the spin-coating process. Black wire: P3HT polymer chain; Large black dots: PCBM; blue dots: DCB molecules; and red dots: OT molecules. a–c) correspond to three stages in the spin-coating process when DCB is the sole solvent; d–f) correspond to three stages in the spin-coating process when OT is added into DCB. Note the difference of PCBM distribution in the final stage of each case, c) and f).

Vertical-phase separation in BHJ

- A vertical-phase separation is usually observed in P3HT and other polymers system due to an inhomogeneous distribution of the donor and acceptor materials inside the BHJ.
- Depending on the solvent-evaporation rate and the substrate surface properties (wetting, surface energy etc.), vertical-phase separation with either donor-rich region or acceptor-rich region near the surface can be tuned.





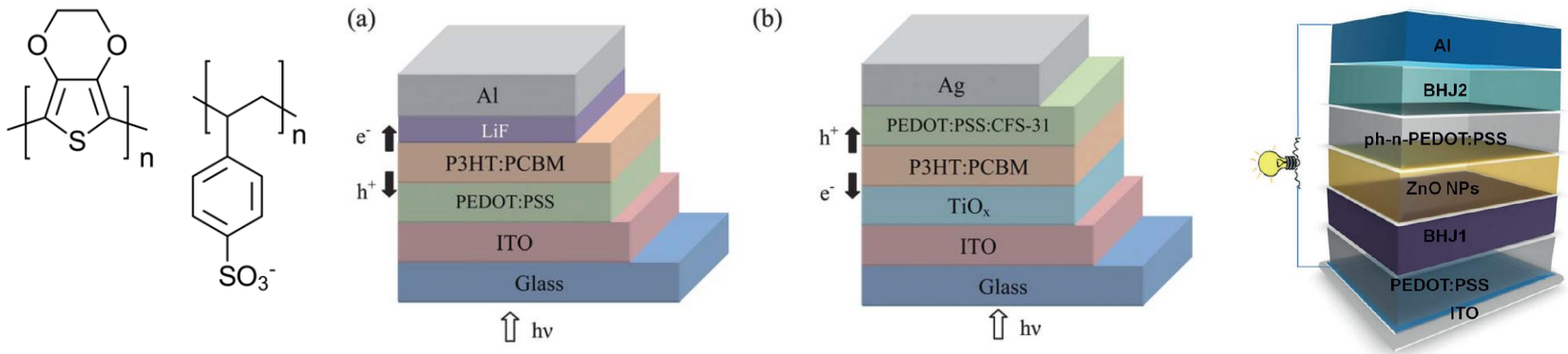
Vertical composition profiles in P3HT:PCBM films as deduced using ellipsometry. a–c) PCBM concentration profiles obtained from analysis of ellipsometric data for P3HT:PCBM blend films: a) spin-coated on fused silica before (blue) and after (red) thermal annealing; b) spin-coated on PEDOT: PSS-coated fused silica before (blue) and after (red) vapor annealing; c) spin-coated on fused silica (left) and on a Si wafer (with native oxide) precoated with a hydrophobic self-assembled hexamethyldisilazane monolayer (SAM).

Interfacial layers in OPV

- While the BHJ is important for charge separation and transport, the interfacial is also crucial for charge collection and exciton blocking. Hence, interfacial layers also play an important role in achieving a high efficiency and stability of OPV.
- Various types of interfacial layers used in OPV
 - ☐ Conductive polymer
 - ☐ Transition metal oxides
 - ☐ Alkali-metal compounds
 - ☐ (Ultra-thin) low-work-function (LWF) metals
 - ☐ Self-assembly monolayers (SAM)

Conductive polymer

- The most commonly used polymer as interfacial layer is poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS). It is a polymer mixture of two ionomers.
- One component poly(3,4-ethylenedioxythiophene) or **PEDOT** is a conjugated polymer and carries positive charges and is based on polythiophene.
- The other component in this mixture is made up of sodium polystyrene sulfonate which is a sulfonated polystyrene (PSS). Part of the sulfonyl groups are deprotonated and carry a negative charge. It is used as counterions to balance the charge neutrality in the system as well as to make the PEDOT water dispersible.
- PEDOT:PSS can be used as hole transport layer in both conventional structure and inverted structure. It can also be used as intermediate layer in tandem cell.



Transition metal oxides

Transition metal oxides offer a unique opportunity to fabricate the devices with large area and at a low cost via solution-processing, which is similar to the solution-processing of organic materials. Depending on the type of metal oxides, they can be used as electron transport layer or hole transport layer.

- Electron transport layer (ZnO , TiO_x)
- Hole transport layer (MoO_3 , V_2O_5 , WO_3 , NiO)

ZnO

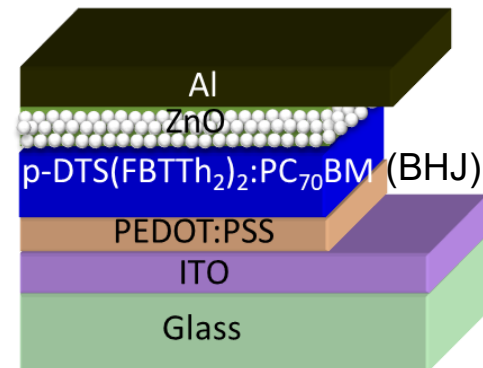
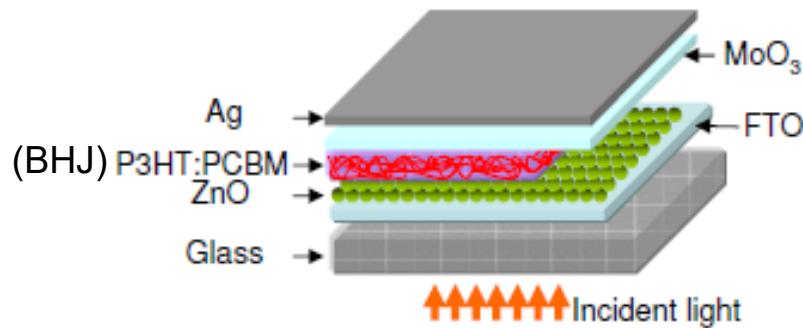
- ZnO is suitable to collect electrons as well as block holes from the BHJ. For example, In P3HT:PCBM system, its conduction band of -4.1 eV is favourable to collect electron from LUMO of PCBM (-3.7 eV) while its valence band of 7.5 eV effectively blocks the hole from HOMO level of P3HT (-6.1 eV).
- Both solution process (sol-gel method and nanoparticle method) and vacuum process (atomic layer deposition) are possible to deposit ZnO.
- Sol-gel method
 - ☐ simple method (mixing precursor in appropriate solvent), no vacuum system is required
 - ☐ need annealing (170 -200 °C to crystallize the film for optimum performance); not suitable to deposit on BHJ in conventional structure
 - ☐ To ensure to form pinhole-free and smooth film, which is possible by tuning the precursor concentration)

- Nanoparticle method

- ☐ No vacuum system is required
- ☐ Preparation of nanoparticle solution is not as simple as sol-gel solution
- ☐ Annealing is not necessary; suitable to deposit on BHJ in conventional structure

- Atomic Layer Deposition

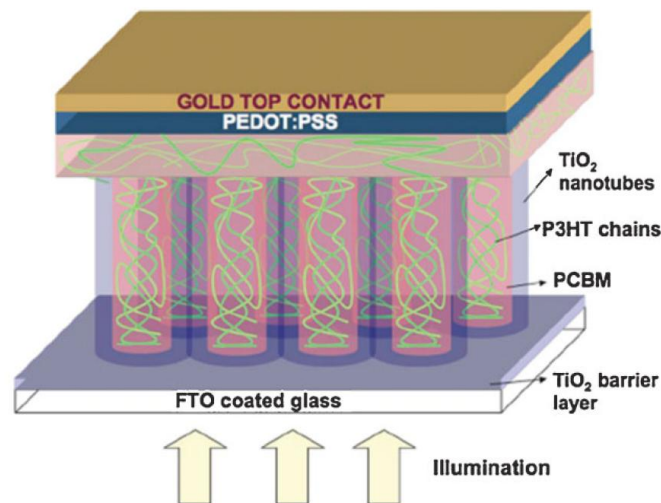
- ☐ Render the best film quality; high crystallinity and pinhole-free
- ☐ Vacuum process; high cost



ZnO electron transport layer used in inverted structure (left) and in conventional structure (right)

TiO_x

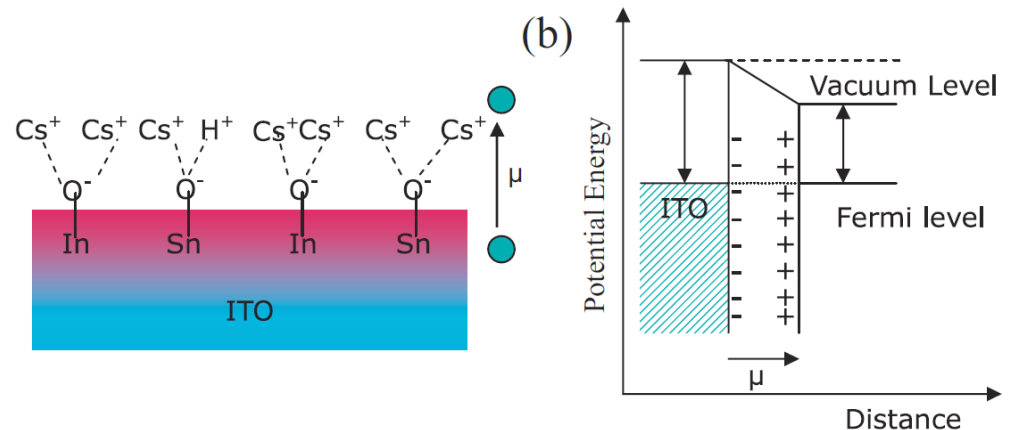
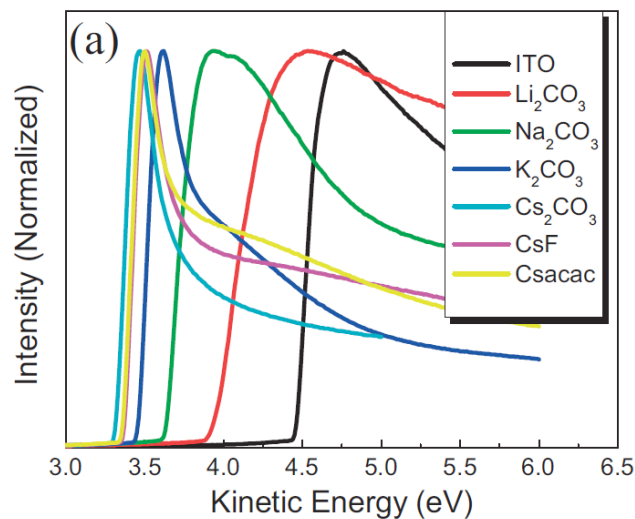
- Deposited by solution-processed
- Similar to ZnO, TiO_x also collect electrons and block holes. Its conduction band (-4.4 eV) is well matched with LUMO of PCBM (-4.3 eV) and its valence band (-7.5 eV) effectively blocks the hole collection to ITO.
- TiO₂ nanotube is also demonstrated as electron transport layer in OPV. The interface between the nanotube and polymer forms additional heterojunction for charge separation. Therefore, this type of structure is called **double-heterojunction** device. Despite additional heterojunction, a significant efficiency improvement is not observed. The efficiency was limited by the poor crystallization and low mobility of polymer in the nanotube as well as insufficient infiltration of polymer in the nanotubes.



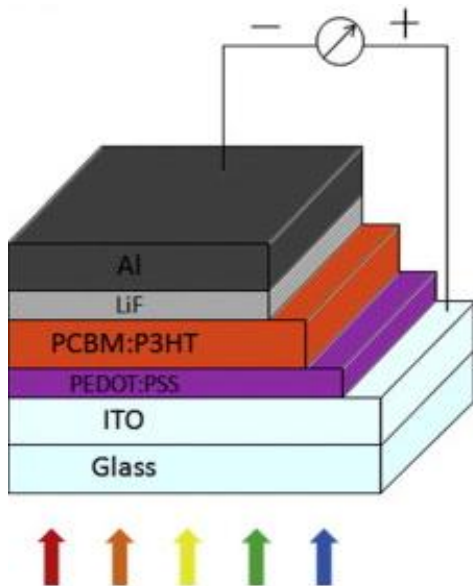
Alkali-metal compounds

- Alkali and alkali-earth metal compounds are considered as candidates for electron injection materials due to their LWF and insensitivity to moisture and oxygen.
- Alkali metal (Li^+ , Na^+ , K^+ , Rb^+ , and Cs^+) acetates and fluorides have revealed the enhanced electron injection in OLEDs.
- Cesium carbonate (Cs_2CO_3) is firstly introduced by Canon to use in OLED but it is also applicable as electron transport layer in OPV. It is easier to handle in comparison with other commonly used alkali-metals.
- Cs_2CO_3 can be deposited by both solution-processing and thermal evaporation, without significant influence on the device performance.
- In inverted OPV, the same metal salt (Cs) with different anions (carbonate, fluoride, and acetate) results in almost the same device performance because buffer layers containing the same metal salt have similar work functions. However, different metal salts with the same CO_3 anion yield different device characteristics owing to different work functions
- A strong dipole layer is formed at the interface of ITO and alkali metal due to the **electron donating ability of metal species**, which alters the work function of ITO.
- As shown in the illustration, the direction of the dipole moments points from the ITO surface to vacuum, and hence reduces the work function of the ITO surface.

- The degree of work-function reduction is determined by the magnitude of the dipole moment, which correlates to the V_{OC} variation, and is proportional to the electron-donating ability of the alkali-metal ions.
- Among the alkali-metal compounds (Cs, K, Na, Li metal with CO_3 anions), Cs_2CO_3 reduces the work function the most due to formation of the strongest dipole moment, leading to higher J_{sc} and V_{oc} and a lower series resistance, compared to other compounds.
- Higher J_{sc} and lower series resistance are related to the enhanced interfacial charge transfer rate at the cathode and higher V_{oc} corresponds to the lower work function of cathode before Fermi level pinning.

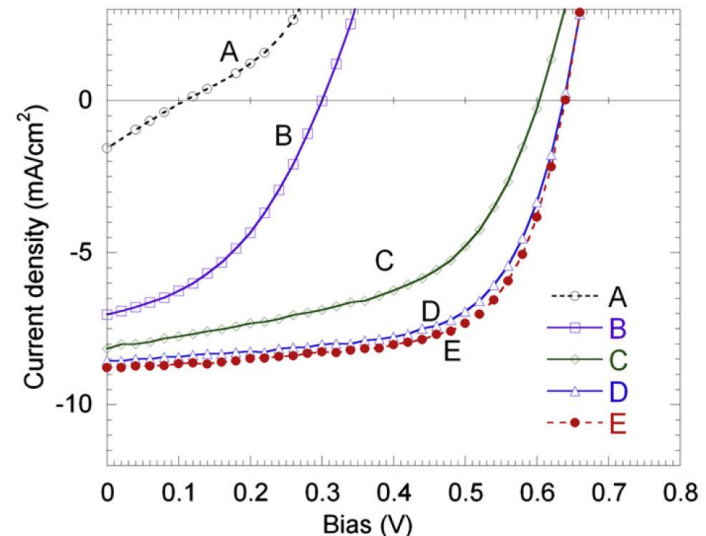
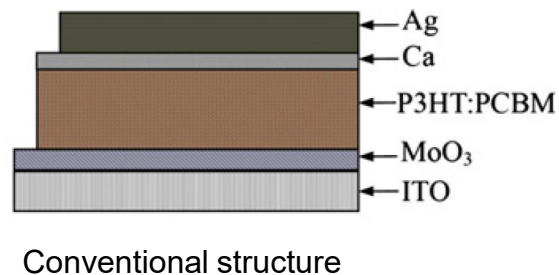
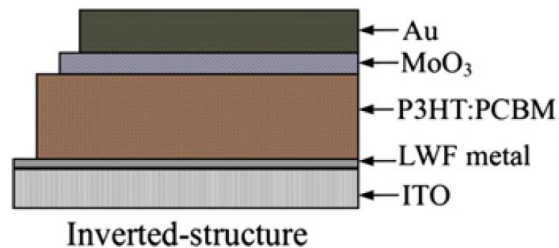


- LiF is another alkali-metal compound, which is commonly used as cathode interfacial layer in conventional structure OPV.
- However, it is not usually used in inverted structure because of a poor performance.



(Ultra-thin) low-work-function metal

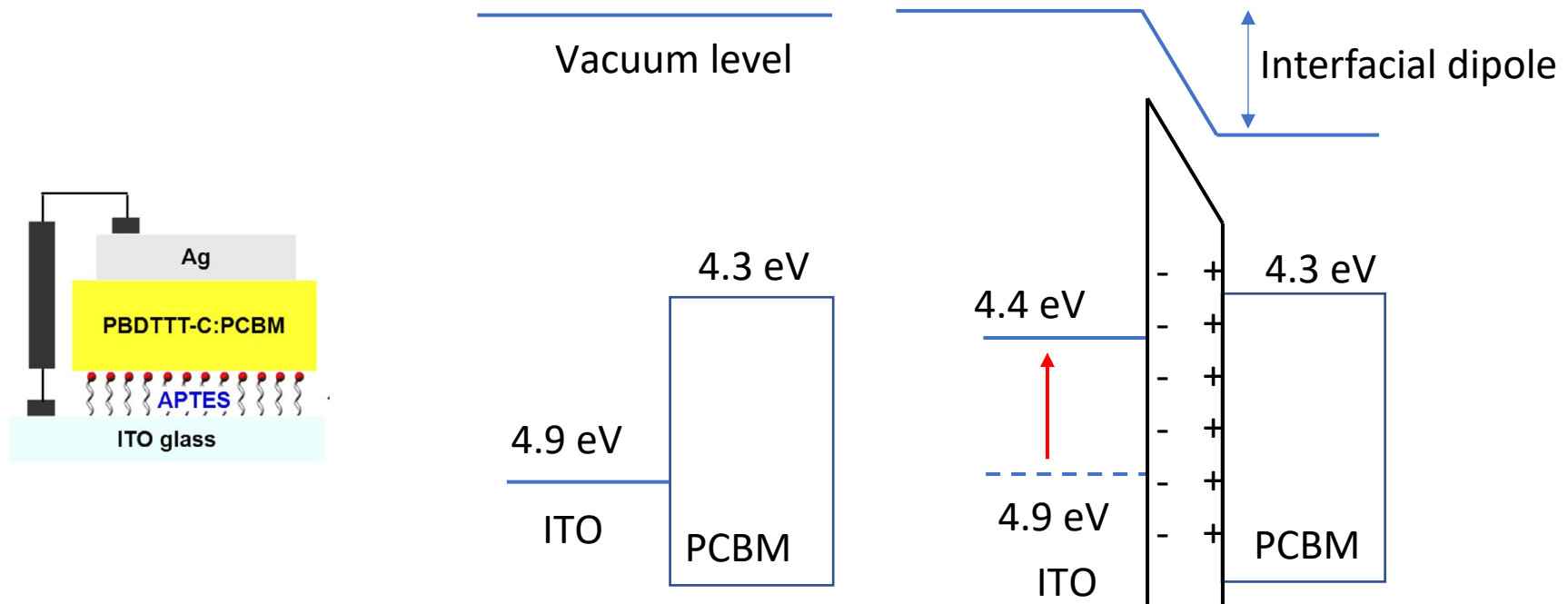
- LWF metals such as Ca, Mg, Ba (Barium) are inserted between BHJ layer and top electrode in conventional structure OPV to enhance the charge collection and built-in potential of the device.
- Similarly, being sandwiched between an ITO electrode and BHJ layer in an inverted cell, LWF metals can serve as an interfacial layer that efficiently collects electron.
- However the metal layer must be ultra thin (1-2 nm) so that it is still transparent.



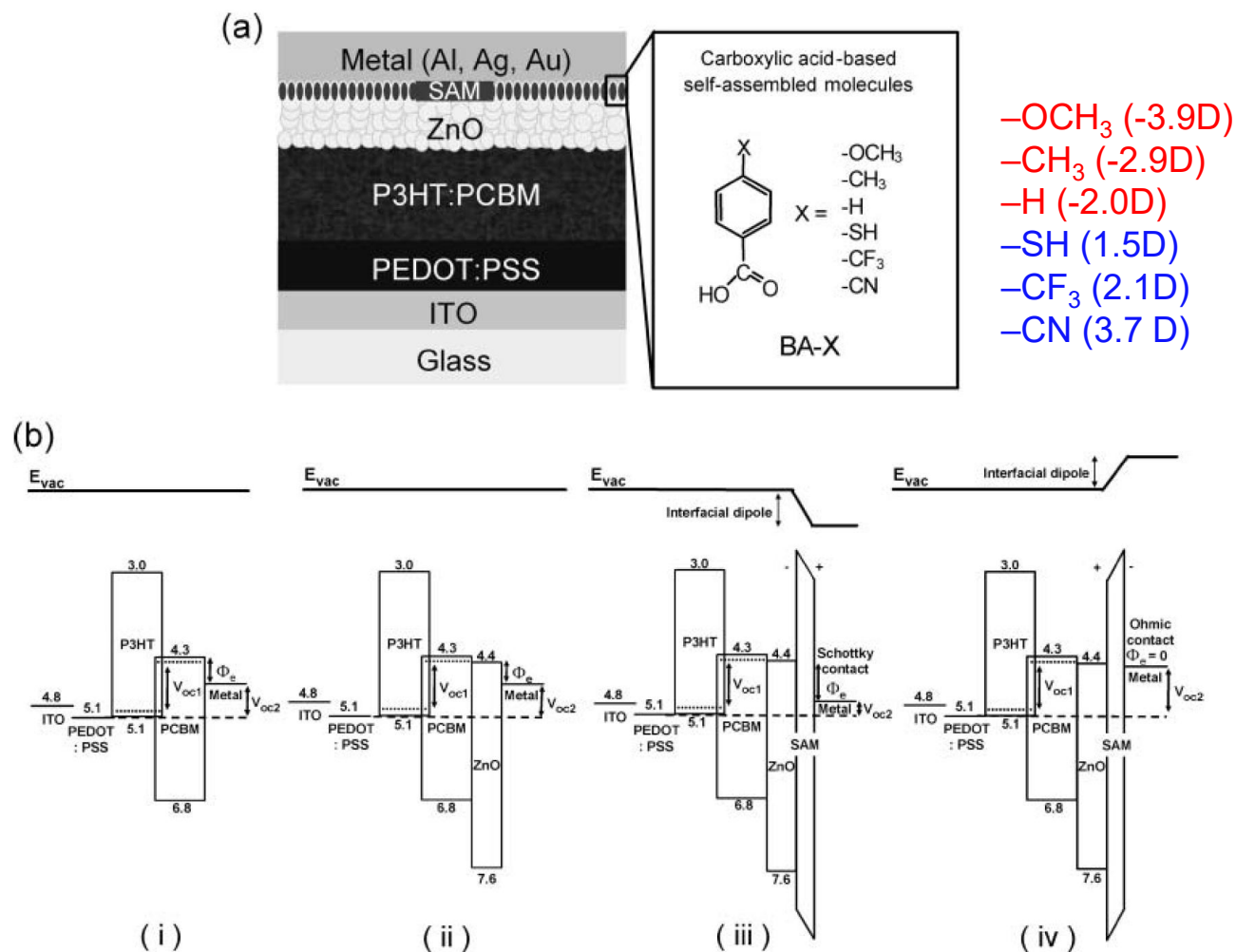
A = ITO, B = Ag, C = Al, D = Mg, E = Ca

Self-assembly monolayers (SAM)

- SAM such as 3-aminopropyltriethoxysilane (APTES) and 1,1,1,3,3,3-hexamethyldisilazane (HMDS)
- Amine groups in APTES and methyl groups in HMDS are electron donating in nature.
- Hence, when APTES and HMDS are deposited on ITO, a dipole moment point away from ITO is created, lowering the work function of ITO .

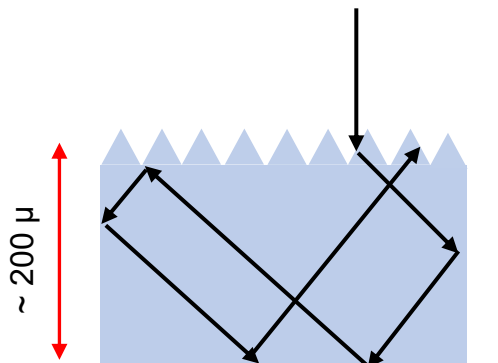
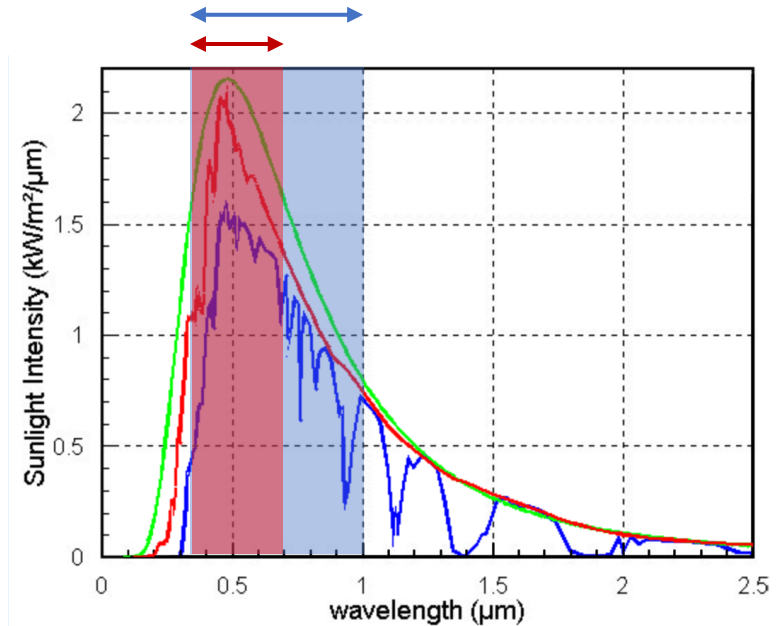
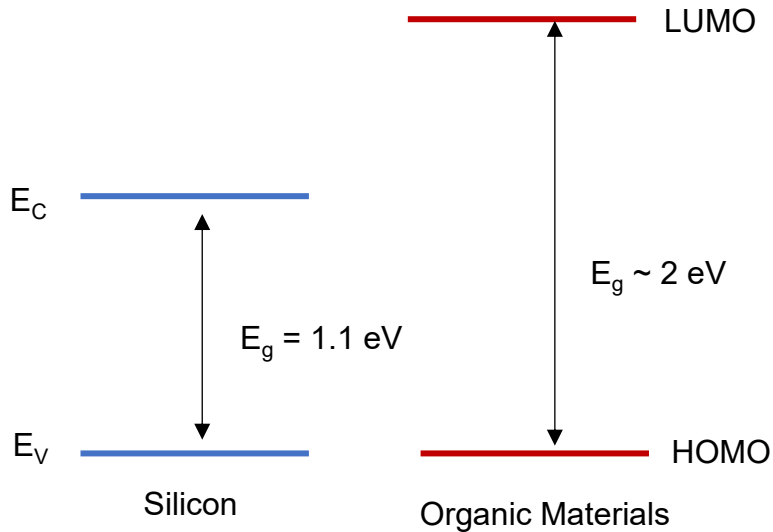


- Carboxylic acid-based SAM

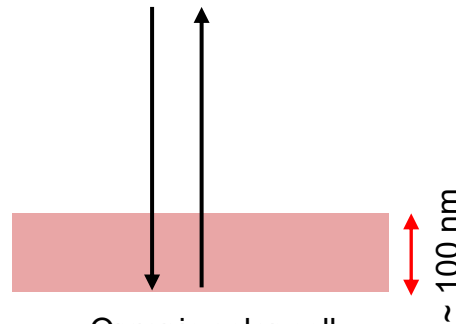


(i) a metal-only cathode; (ii) a ZnO/metal bilayer cathode; (iii) a ZnO/SAM/metal cathode with an interfacial dipole directed towards the metal surface (positive dipole); and (iv) a ZnO/SAM/metal cathode with an interfacial dipole directed away from the metal surface (negative dipole)

Light manipulation in OPV



Silicon solar cell
Long optical absorption length



Organic solar cell
Short optical absorption length

Absorption Coefficient
 Silicon $\sim 10^4 \text{ cm}^{-1}$
 Organic Material $\sim 10^5 \text{ cm}^{-1}$

Why light manipulation is important in OPV?

- Optical Path Length

- ❖ Although the absorption coefficient of organic material is one order of magnitude higher than silicon, the thickness of the light absorption layer in OPV is limited to ~ 100 nm (due to the low mobility and short diffusion length of organic materials). The thickness of light absorption layer is ~ 200 μm in bulk silicon solar cell.
- ❖ Moreover, the optical path length in silicon solar cell is about 50 times of the thickness of the light absorption layer due to light trapping via total internal reflection.
- ❖ The optical path length in OPV is only two times of the thickness of the light absorption layer.

- Spectral coverage

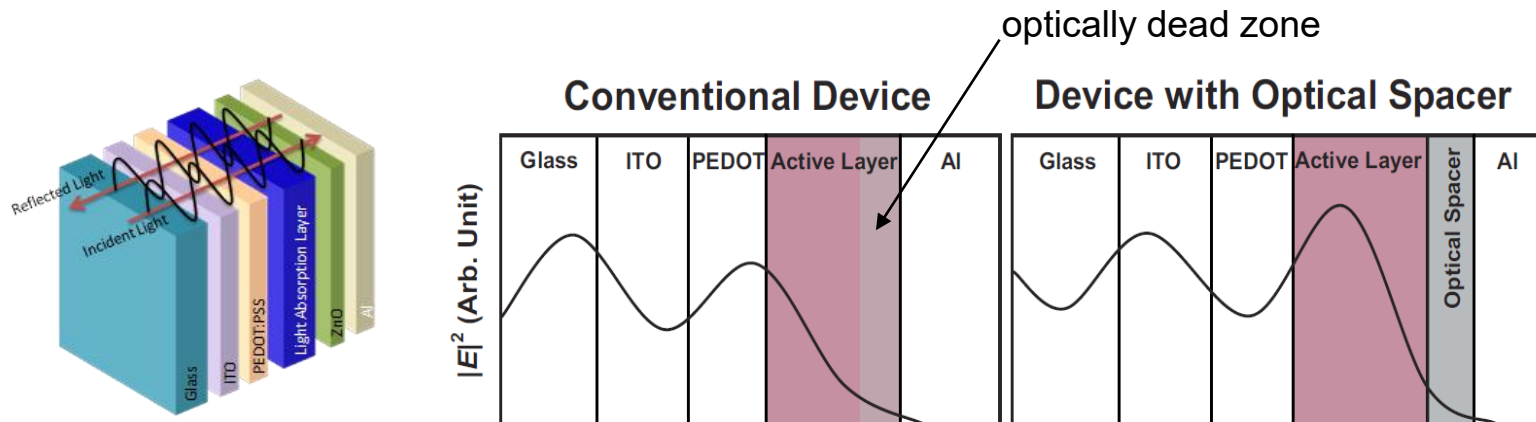
- ❖ OPV usually cannot absorb light in NIR region due to the large bandgap of material.
- ❖ The bandgap of commonly used organic materials in OPV is between 1.5 eV to 2.0 eV, which is much higher than that of silicon (1.1 eV). (Note: bandgap as low as 1.0 eV has been synthesized but those very low bandgap polymers do not give high efficiency for solar cell)

Techniques to increase light absorption in OPV without increasing the thickness

- Optical spacer
- Surface plasmon resonance
- Light trapping

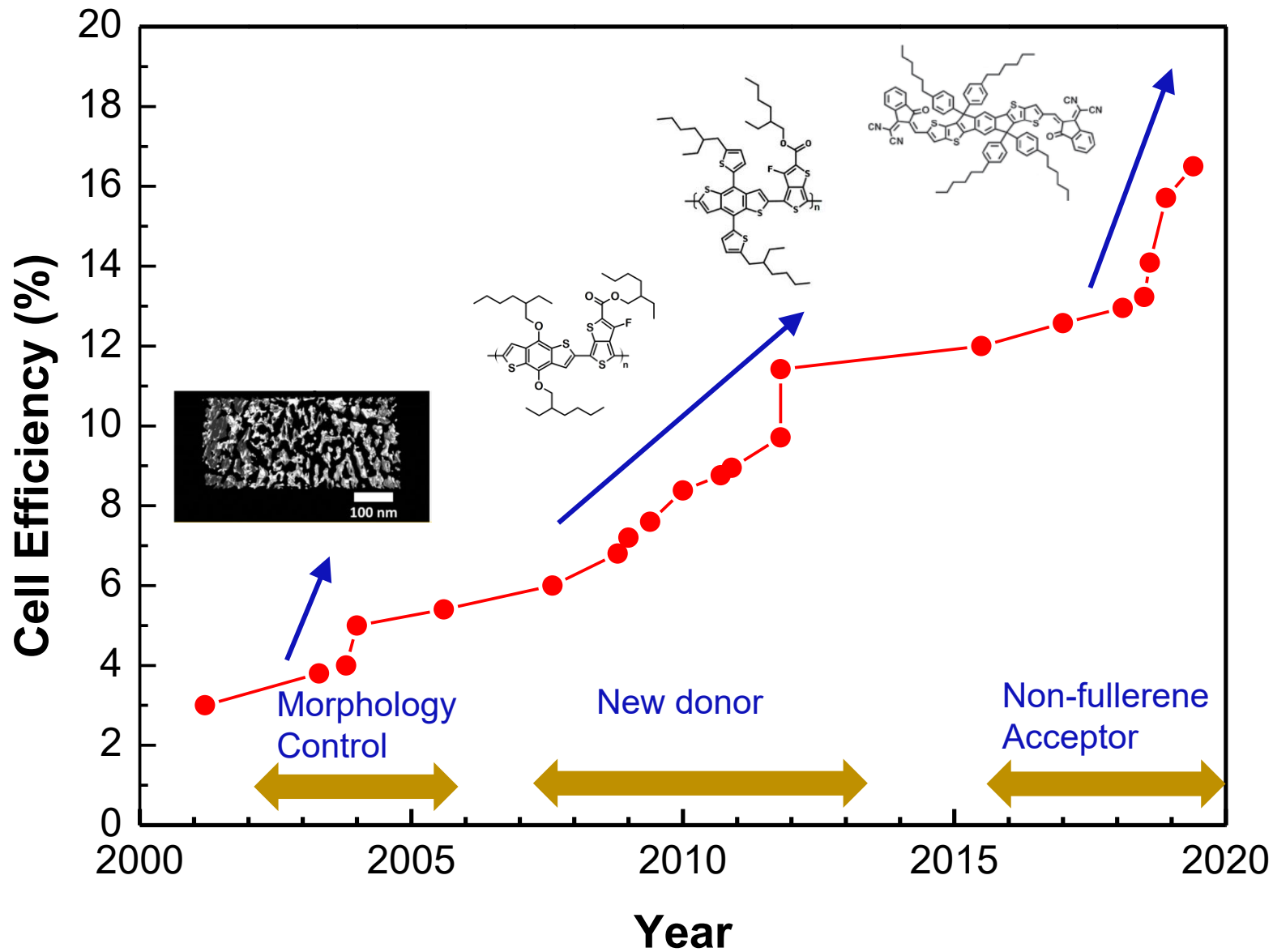
Optical Spacer (Optical Microcavity Resonance)

- Because of optical interference between the incident (from the ITO side) and back-reflected light, the intensity of the light is zero at the metallic (Al) electrode.
- This “optical interference effect” is especially important for thin-film structures where layer thicknesses are comparable to the absorption depth and the wavelength of the incident light.
- A schematic representation of the spatial distribution of the squared optical electric-field strength is shown below.



- Without optical spacer, part of the active layer (BHJ) is under the “optically dead zone”.
- the insertion of an optical spacer between the active layer and the metal electrode can place the active layer in a more favorable region of optical electric field.
- Good candidate for optical spacer: ZnO , TiO_2

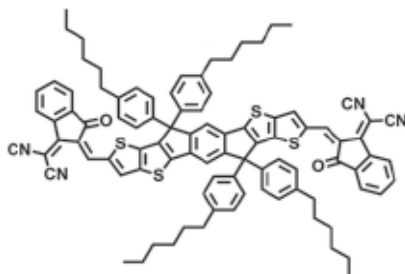
- Further reading for Surface plasmon resonance and light trapping in OPV
- “Light Manipulation in Organic Photovoltaics”, Advanced Science, (2016), Vol 3, Page 1600123,
- “Plasmonic-enhanced Organic Photovoltaics: Breaking the 10% Efficiency Barrier”, Advanced Materials, (2013), Vol 25, Page 2385.



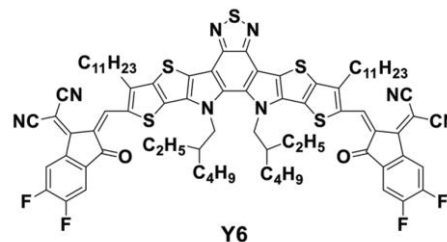
Non-fullerene Acceptor

- What is non-fullerene acceptor?

Electron-acceptor based on small molecules



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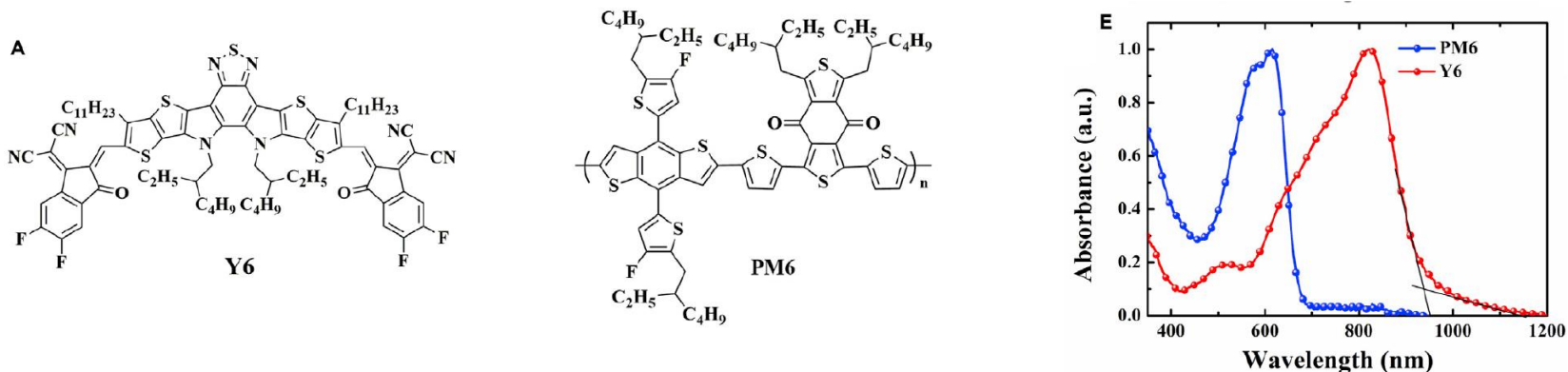


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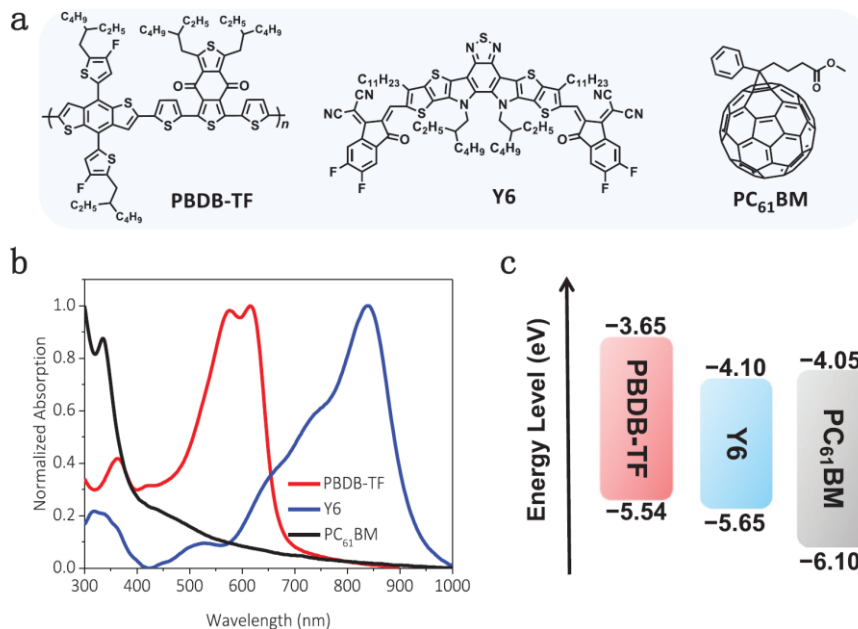
Y6

- Why non-fullerene acceptor?
- Fullerene acceptor barely absorbs light in visible region (absorbs only UV and near UV region)
- Non-fullerene acceptor has a strong absorption in visible region
- The absorption spectrum can be tuned to complement with that of donor

- Over 15% efficiency with non-fullerene acceptor



- Even higher efficiency (over 16%) with Ternary Blend

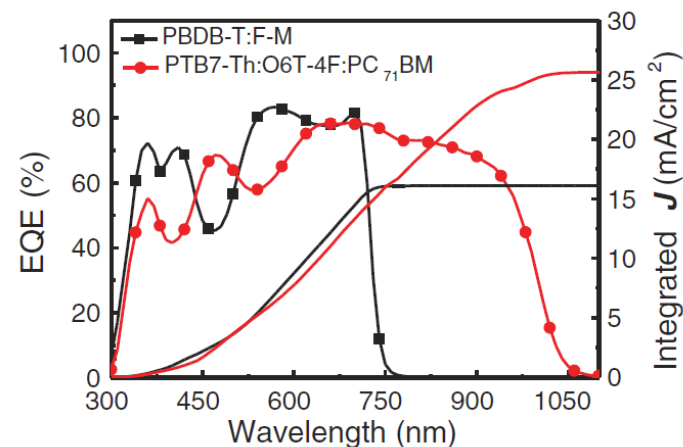
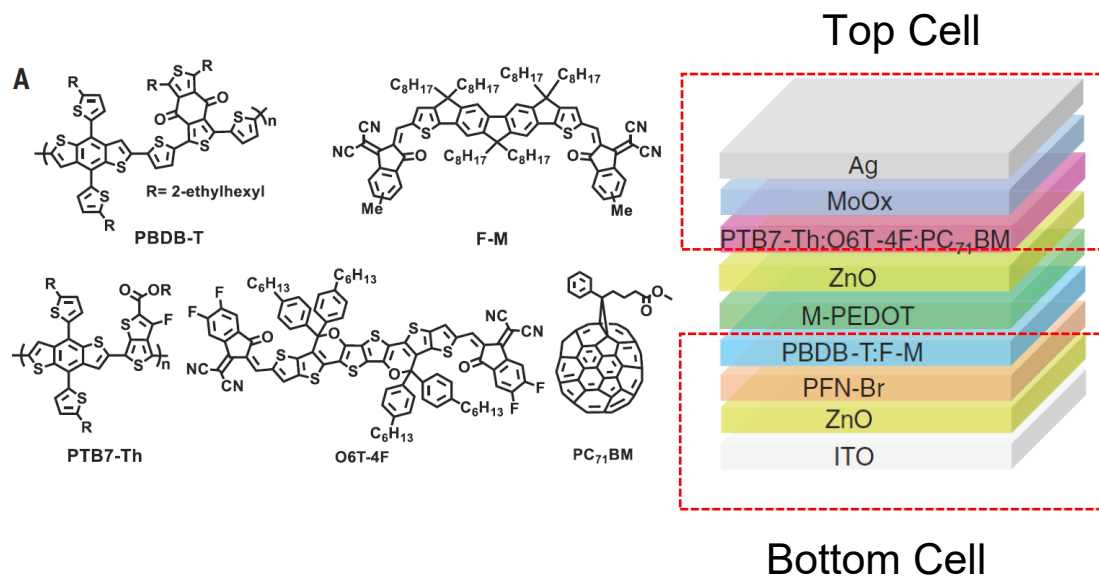
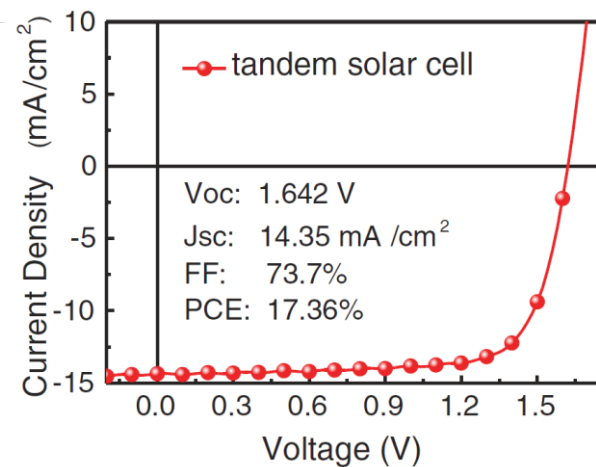


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Adv. Mater. **2019**, 1902302

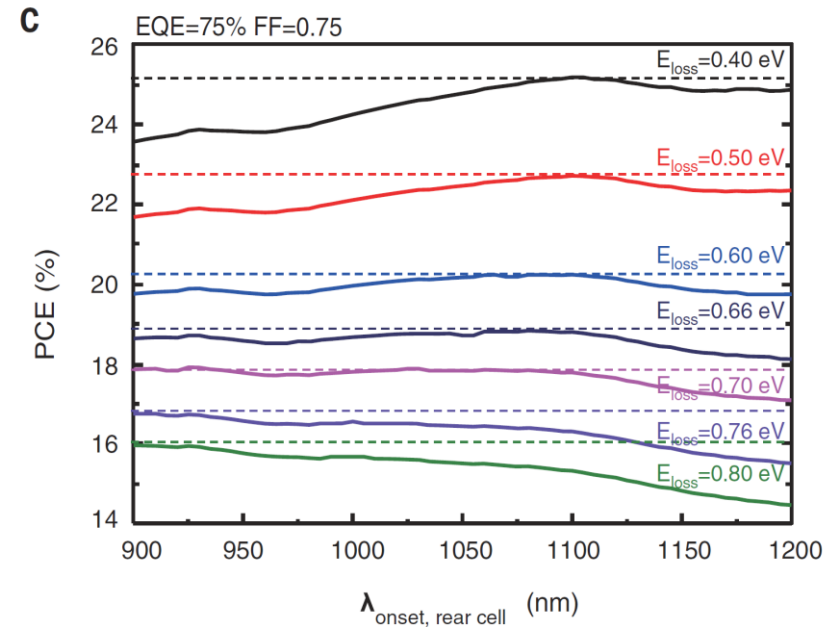
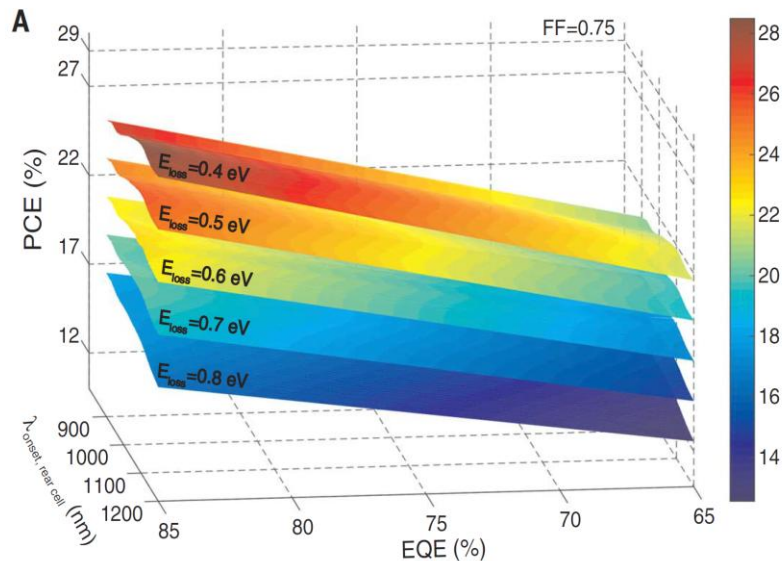
Organic and solution-processed tandem solar cells with 17.3% efficiency

Lingxian Meng¹, Yamin Zhang¹, Xiangjian Wan^{1*}, Chenxi Li¹, Xin Zhang¹, Yanbo Wang¹, Xin Ke¹, Zuo Xiao², Liming Ding^{2*}, Ruoxi Xia³, Hin-Lap Yip³, Yong Cao³, Yongsheng Chen^{1*}



Meng et al., Science, 2018, 361, 1094

- Efficiency 17.3% is not the limit!
- Semi-empirical predication
 - Single junction: 18%
 - Double junction: 25%
 - Triple junction: 28%



Meng et al., Science, 2018, 361, 1094